

Explainable artificial intelligence models in intrusion detection systems

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ABSTRACT

This study introduces the growing necessity of Explainable Artificial Intelligence (XAI) in the context of cybersecurity, particularly in Intrusion Detection Systems (IDSs). As Artificial Intelligence (AI) technologies, especially complex neural networks, become more prevalent and difficult to understand, the opacity of these models poses increasing risks in critical fields like cybersecurity. XAI aims to address these concerns by making AI models and their outputs more interpretable and transparent, thus fostering trust and secure use. Despite the rising interest and application of XAI in various domains, its integration into IDSs remains insufficient. This research provides a comprehensive overview of XAI, examining its definitions, terminologies, and the evolution of the field. It delves into the opportunities, challenges, and research needs within XAI, as well as the latest developments, tools, and technologies for implementing XAI in AI-based IDSs, alongside associated risks. The study also summarizes and demonstrates a gap in the literature regarding comprehensive reviews of IDS solutions that incorporate XAI, aiming to fill this gap by detailing the application of XAI in IDS research. The findings are expected to guide researchers, experts, users, and policymakers in the domains of AI, XAI, and IDS.

1. Introduction

There have been various security approaches and technologies to ensuring cyber and information security. While cryptographic approaches are generally used to protect information and provide confidentiality and integrity, many different security approaches and technologies such as firewalls, antiviruses, Denial of Service (DoS) mitigation solutions, and intrusion detection systems (IDS), have been used to protect information accessibility. In addition, developing new tools using the most up-to-date techniques and configuring them according to customized rules and policies helps us prevent attacks to a large extent and ensure information security more. IDSs are software and hardware that enable us to take precautions against attacks on information systems, detect attacks, and report them by generating warnings or alarms. By using IDSs, unauthorized access and misuse of systems are detected and the damages they may cause are prevented as early and automatically as possible. IDSs have reached a market size of 5.6 billion dollars in 2022. The market size is estimated to reach 15.5 billion dollars, growing by over 11.5% every year until 2028 ([Market Data Forecast](#)). While IDS solutions had a share of over 64% in the market in 2018, IDSs are expected to dominate the market by 2025

([Global Market Insights](#)). Having various advanced features such as data loss prevention, unauthorized access prevention, malicious activity detection, web application firewall and DoS prevention have enabled the market for IDSs to grow.

IDSs, especially equipped with developing technologies such as artificial intelligence (AI) and deep learning (DL), are becoming an important actor day by day in ensuring the security of information communication technology (ICT) systems. IDSs that use AI techniques, also referred to as intelligent IDSs, significantly contribute to ensuring cybersecurity as an increasingly popular and effective defense mechanism.

There has been a significant increase in the popularity and use of AI and ML over the past decade. AI continues to be used more widely and develops in many different areas of daily life. A study by [Grand View Research](#) evaluated the size of the global AI market at US\$136.55 billion in 2022 and predicted a compound annual growth rate (CAGR) of 37.3% from 2023 to 2030 ([Grand View Research](#)). In the research conducted by Next Move Strategy Consulting, the size of the global explainable artificial intelligence (XAI) market was evaluated as 4.4 billion US dollars in 2021 and predicted to reach 21 billion US dollars with a compound a CAGR of 18.4% from 2022 to 2030 ([Next Move Strategy Consulting](#)). In

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parallel with these developments, the use of AI in IDSs is increasing significantly, and as a result, IDSs are developing more and more every day. IDSs are widely used and are becoming an indispensable part of cybersecurity strategies. As a result of this trend, a wide variety of new approaches and techniques have emerged that achieve surprising results, such as DL. These models, which have been introduced in recent years, have significantly improved compared to previous models in terms of prediction performance (Accuracy, Precision, Recall, F1-Score, etc.). However, there are several features of these models that have not received much attention, such as complexity, interpretability, and fairness. The increasing adoption of AI technologies, especially popular, incomprehensible, complex, and opaque ML models such as neural network models, is becoming increasingly difficult to understand. This makes the problem worse and more important in a field such as cybersecurity because trusting a system that cannot explain the reason for important decisions and leaving it alone poses obvious dangers. As a result, the complexity of current techniques is increasing, resulting in AI/ML black box models that cannot be easily examined or verified and are too complex for any human to comprehend. Such models can solve various complex problems, but they remain unreliable due to situations where predictions have negative effects. To overcome this problem, XAI proposes a transition to more interpretable, explainable, and understandable AI. XAI, a holistic approach to solutions, uses various methods to understand, comprehend, and interpret AI models. The field of research in XAI aims to solve the explainability and interpretability problem of AI and explain AI model's decision to users in terms that users can understand. It reveals more details about the behavior of complex ML models that help experts validate model predictions. However, as AI is frequently used in high-risk areas such as cybersecurity, health, finance, transportation, autonomous driving, military, and law, the need for stakeholders to understand AI is increasing due to the dangerous consequences it may cause. This need for understanding also confronts researchers as a problem that must be overcome in order for people to trust and adopt AI technologies.

XAI approaches have been adopted to address the need for explainability in high-risk areas such as cybersecurity (Chen et al., 2021), healthcare (Tjoa and Guan, 2021a), finance (Silva et al., 2022), transportation (Zhang et al., 2020), autonomous driving (Lu et al., 2021), the military (Nguyen et al., 2020), and law (Dorr et al., 2021). Recently, XAI techniques have started to be used as a cybersecurity solution for IDSs, as well as in many other critical research areas. Cybersecurity researchers, users, policymakers, and all stakeholders in this field need to understand the importance of AI in order to fully benefit from its

capabilities. XAI plays an important role in understanding AI models in the field of cybersecurity. The focus of this study differs from other studies as it highlights the latest developments and approaches to developing XAI applications and presents available tools and technologies for related topics and concepts, and provides a summary to elaborate the power of XAI models in IDSs. However, although some other studies exist on XAI and cybersecurity independently, no comprehensive study focuses on a broad review of IDS solutions based on XAI. Therefore, to fill this gap, this article provides a comprehensive review on XAI in IDSs for the last five years, highlights existing state-of-the-art work, the need for XAI, and the importance of XAI for IDSs.

1.1. Related works

Few studies on XAI have been published in the literature. Particularly in the field of cybersecurity, very few studies have been conducted. However, XAI and cybersecurity have been addressed separately in many studies. Additionally, there is no comprehensive research in the literature specifically focusing on the use of XAI in IDSs. Generally, it has been evaluated as a brief subheading in studies addressing cybersecurity. Fig. 1 shows the number of all XAI studies, XAI studies in the field of cybersecurity, and XAI studies in the field of IDSs in Google Scholar between 2018 and 2024. It can be seen from the figure that there are very few studies that cover both XAI and IDS domains together. For example, in 2023, the percentage of studies that include IDS among all XAI studies is approximately 3%. This study fills this gap in the literature by providing a comprehensive introduction to various XAI techniques applied in IDSs. Our study also offers a thorough background on XAI.

There have been many studies in the literature that aim to guide researchers by examining the use of AI in the field of cybersecurity. For example, Veiga (Veiga, 2018), by detailing the innovative solutions and advantages offered by AI in network security, demonstrate that AI can increase the effectiveness of existing security systems. The study provides important insights for better understanding the potential of AI in network security and for guiding future innovations in this field. In (Wickramasinghe et al., 2018), Wickramasinghe et al. examined the general applications of deep learning (DL) in ensuring the security of cyber-physical systems (CPS). They focused on how deep learning can be integrated into CPS security, existing approaches, and the generalizability of these approaches. As a detailed study on the use of AI in cybersecurity, Shaukat et al. (2020) provide a comprehensive review of ML techniques used in the field of cybersecurity over the past decade. This study explores how ML techniques are applied in cybersecurity and

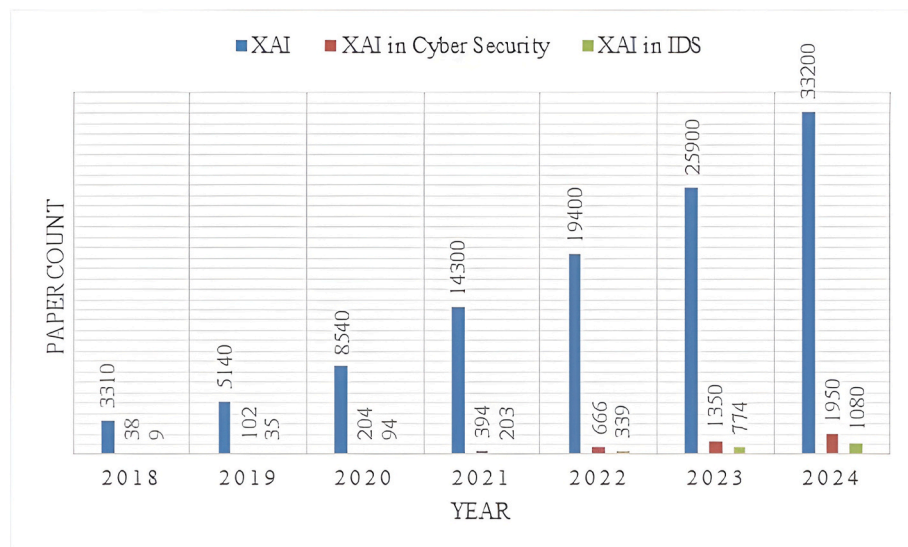


Fig. 1. Number of XAI, XAI in cyber security and XAI in IDS articles published between 2018 and 2024.

the advancements in this field, showing that supervised and unsupervised learning techniques are used in many areas such as intrusion detection, malware detection, and fraud detection.

Due to the significant advances of AI-based IDSs in the field of cybersecurity, there are many studies that focus solely on examining and analyzing the use of AI in IDSs. For example, [Mishra et al. \(2019\)](#) focused on how ML techniques are used in IDSs, which are critical components of network security. The study analyzes in detail how ML techniques, particularly supervised, unsupervised, and semi-supervised learning, can be integrated into IDS systems, as well as the advantages and limitations of these approaches. It is demonstrated that ML methods can be successfully applied in IDSs, and these systems can automate and improve intrusion detection. [Gumusbas et al. \(2021\)](#) provide a comprehensive review of DL methods and databases for cybersecurity and IDS. Although IDSs are not covered in detail in the study, the datasets used in the application of DL methods are discussed more extensively. Another study on the use of AI in IDSs was conducted by [Muneer et al. \(2024\)](#). This study analyzes in detail how AI-based approaches are used in IDSs, emphasizing that AI is a powerful tool for enhancing the effectiveness of IDSs.

Due to the critical nature of decisions in the field of cybersecurity, many studies have emphasized the importance of investigating XAI from a cybersecurity perspective. [Kuppa and Le-Khac \(2021\)](#) examined how XAI methods can be used to increase the reliability of AI systems and how defenses can be developed against adversarial attacks. The study reveals the potential of XAI in cybersecurity while also highlighting the current challenges and future improvements in this area. Another study presented in [\(Srivastava et al., 2022\)](#) comprehensively covers the current status of XAI in cybersecurity, the challenges encountered, unresolved issues, and future research directions. [Capuano et al. \(2022\)](#) emphasize the importance of XAI in cybersecurity and analyze the existing techniques, their application areas, advantages, and limitations. The study also provides suggestions for future research to further promote the use of XAI in cybersecurity. Similarly, [Fabien et al. \(2022\)](#) conducted a comprehensive literature review on XAI in cybersecurity, examining how XAI is used in AI-based cybersecurity systems, its advantages, challenges, and future research directions. Another study on the use of XAI in cybersecurity was conducted by [Zhang et al. \(2022\)](#). This study provides an overview of the current state of research in this field, discussing the XAI techniques used in cybersecurity, the advantages these techniques offer, and the challenges they face. It also provides suggestions on how XAI can be further developed in cybersecurity. Like the aforementioned studies, [Rjoub et al. \(2023\)](#) provide a comprehensive discussion on the current state of XAI in cybersecurity and the latest developments in this field.

Due to the increasing need for explainability in many different critical application domains, numerous studies have been conducted that conceptually address explainability in a comprehensive manner and examine it in depth. In his study, [Rudin \(2019a\)](#) demonstrates why it is critical to prefer explainable models over black-box models and argues that these models should especially be preferred in high-risk decision processes by proposing approaches based on explainable models. He states that explainable models provide transparency, trust, and accuracy, while also facilitating auditing and error correction processes. [Sahakyan et al. \(2021\)](#) presented a comprehensive review on the use of XAI methods, particularly for tabular data. The study examined existing XAI techniques for tabular data, analyzing which methods were used, along with their advantages and limitations. [Alicioglu and Sun \(2021\)](#) presented a comprehensive review on how XAI methods are supported by visual analytics. The study explores visual analytics techniques integrated with XAI methods, analyzing the advantages of these techniques, the challenges encountered, and potential future research areas. [Ahmed et al. \(2022\)](#) conducted a literature review examining how to make AI applications in Industry 4.0 explainable. It details which XAI techniques can be used to transform AI applications in Industry 4.0 to XAI, how they can be implemented, and the opportunities in this area.

[Kok et al. \(2022\)](#) provided a comprehensive literature review on how XAI methods can be applied to make AI models used in the Internet of Things (IoT) more transparent. The study highlights the challenges XAI faces in IoT and suggests future research directions in this area. [Chamola et al. \(2023\)](#) presented a comprehensive review on Trustworthy and Explainable Artificial Intelligence. The study outlines why XAI is critical in the process of creating trustworthy AI, the components of trustworthiness, and how XAI can be used in AI systems. It particularly emphasizes the role of XAI in providing transparency, security, and fairness in fields such as healthcare, finance, law, and autonomous systems. [Lettrache and Ramdani \(2023\)](#) conducted a comprehensive literature review and case study on model-agnostic XAI methods. The study introduces model-agnostic XAI methods, explains how these methods work, and examines their applicability to different AI models. It also demonstrates how these techniques can be applied in real-world scenarios through a case study.

[Table 1](#) summarizes how our work is more comprehensive than other survey and review articles in the literature and highlights our contributions to these studies. The analysis of previous studies reveals our motivation and contributions to this research. While existing studies have analyzed AI applications in cybersecurity in detail, some have only examined XAI methods for a narrow set of cybersecurity applications. Additionally, some studies have not provided a detailed explanation of the background of XAI and cybersecurity. Upon examining [Table 1](#) and it can be seen that this study is unique and distinct in terms of its detailed analysis of XAI and its comprehensive exploration of the relationship between IDS and XAI, compared to previously published studies in the field.

1.2. Black box problem

Black box models are ML models that cannot be understood by just looking at their parameters. Unlike linear models and decision trees, ML and especially DL models are often described as black box models due to their complexity. Black box models are inherently opaque. It can be difficult for even skilled experts to fully grasp these 'black box' models. Many ML and DL models essentially appear as "black boxes," as shown in [Fig. 2](#), which do not specify the details of decision in their internal mechanisms and predictions.

This lack of transparency and understanding in ML and DL algorithms can have serious consequences for trust and adoption in the models. For example, how can we know whether model predictions are wrong? This is especially vital in high-risk areas such as healthcare and cybersecurity. If a trained model has 99% accuracy, can a cybersecurity professional trust its attack prediction? What if the model misses the most important attack events? What if the high accuracy is due to data leakage in the test data, and out-of-sample performance is much worse? Because of questions like these, XAI is vital to the adoption of ML. It becomes necessary for ML models to be explainable for high-risk decisions such as finance, medical diagnosis, and cyber security ([Kleinberg et al., 2018](#); [Lakkaraju et al., 2019](#)). The usefulness of black box models is becoming increasingly questionable, and some researchers argue that they should not be used for high-risk decisions ([Rudin, 2019b](#)).

1.3. Contributions

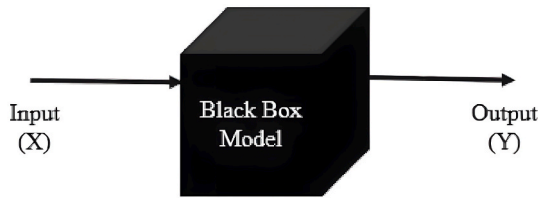
This study fundamentally provides a holistic perspective on different aspects of XAI. It is a guide that shows those who are new to XAI or experts in the field what XAI is, what it aims for, what it can do, what its current situation is, where current trends, terms, and techniques can be interpreted effortlessly and aims to provide significant gains to those who are interested. However, after presenting general information about IDSs, a comprehensive study is presented that focuses on reviewing, classifying, and summarizing a wide range of IDS solutions based on XAI. In summary, this study contributes to researchers, experts, users, and policy makers in the field of cybersecurity, AI, and XAI as

Table 1

Comparison of existing surveys with our work.

Reference	Year	XAI				Cyber Security	IDS			
		Taxonomy	Techniques	Challenges	Evaluation		Taxonomy	Techniques	Challenges	Evaluation
Veiga (2018)	2018	N/A	N/A	N/A	N/A	✓	N/A	N/A	N/A	N/A
Wickramasinghe et al. (2018)	2018	N/A	N/A	N/A	N/A	✓	N/A	N/A	N/A	N/A
Mishra et al. (2019)	2019	N/A	N/A	N/A	N/A	✓	N/A	✓	✓	✓
Rudin (2019a)	2019	N/A	N/A	✓	✓	N/A	N/A	N/A	N/A	N/A
Shaukat et al. (2020)	2020	N/A	N/A	N/A	N/A	✓	N/A	✓	✓	✓
Gumusbas et al. (2021)	2021	N/A	N/A	N/A	N/A	✓	N/A	✓	✓	✓
Kuppa and Le-Khac (2021)	2021	N/A	✓	✓	✓	✓	N/A	N/A	N/A	N/A
Sahakyan et al. (2021)	2021	✓	✓	✓	✓	N/A	N/A	N/A	N/A	N/A
Alicioglu and Sun (2021)	2021	✓	✓	✓	✓	N/A	N/A	N/A	N/A	N/A
Ahmed et al. (2022)	2022	✓	✓	N/A	✓	✓	N/A	N/A	N/A	N/A
Srivastava et al. (2022)	2022	N/A	✓	✓	✓	✓	N/A	N/A	N/A	N/A
Kok et al. (2022)	2022	✓	✓	✓	✓	✓	N/A	N/A	N/A	N/A
Capuano et al. (2022)	2022	✓	✓	✓	N/A	✓	✓	✓	N/A	✓
Fabien et al. (2022)	2022	✓	✓	✓	✓	✓	N/A	N/A	N/A	N/A
Zhang et al. (2022)	2022	✓	✓	✓	✓	✓	N/A	✓	N/A	N/A
Chamola et al. (2023)	2023	N/A	✓	✓	✓	N/A	N/A	N/A	N/A	N/A
Rjoub et al. (2023)	2023	✓	✓	✓	✓	✓	N/A	N/A	N/A	✓
Letrache and Ramdani (2023)	2023	✓	✓	N/A	✓	N/A	N/A	N/A	N/A	N/A
Muneer et al. (2024)	2024	N/A	N/A	N/A	✓	✓	✓	✓	N/A	✓
Present Study	2024	✓	✓	✓	✓	✓	✓	✓	✓	✓

N/A: Not Applicable.

**Fig. 2.** Black box model.

follows.

- 1) This study first provides an up-to-date and comprehensive analysis that emphasizes the importance of XAI for achieving trustworthy AI. It also introduces several key concepts, including a detailed definition of XAI, its terminology, objectives, current XAI toolboxes, and a thorough classification of XAI techniques, offering researchers a holistic perspective. In particular, it clarifies the interrelated concepts used in the literature as alternatives to explainability, which often lead to confusion.
- 2) Important criteria for evaluating the quality and meaningfulness of explanations produced by XAI techniques are presented, as well as indicating current research challenges, open problems and possible solutions.
- 3) The study rationalizes the motivations for integrating XAI into AI-based IDS models and reveals the opportunities, challenges, and further research needs for the use of XAI in IDSs.
- 4) Although there are some other studies in the literature on XAI and cybersecurity, there is no comprehensive study that extensively reviews IDS solutions based on XAI.
- 5) The study also addresses the relationship between the fields of XAI and LLM, which have gained increasing popularity in recent years. Evaluations were made regarding the mutual impact of these two important and evolving areas of study.

1.4. Research methodology

The aim of this study is to provide a comprehensive and up-to-date background on XAI and to examine the current status and studies in the literature on XAI applications in intrusion detection systems. For this purpose, the reference set was created by carefully selecting reputable

articles and studies published in peer-reviewed conferences, journals, preprints, and websites, following the research methodology illustrated in Fig. 3. The following criteria were established for collecting the reviewed research articles.

- 1) First, the basic keywords “XAI” and “IDS” were used in search engines such as Google Scholar ([Google scholar](#)), IEEE Xplorer ([IEEE Xplore](#)), Springer ([SpringerLink](#)), and ScienceDirect ([ScienceDirect](#)). Additionally, search results were obtained using synonymous keywords like “Explainable Artificial Intelligence”, “Explainable AI”, “Explainable ML”, “Interpretable ML”, “Interpretable Artificial Intelligence”, “Interpretable AI”, and “Intrusion Detection System”.
- 2) Using the specified keywords, advanced search methods such as “Explainable AI” AND “Intrusion Detection System” OR “XAI” AND “IDS” were employed in the search engines, and results were obtained. The number of studies retrieved from each search engine is shown in Fig. 4. The selection criteria were defined as addressing both XAI and IDS together and being an IDS design study with explainability features.
- 3) Studies published in between 2018 and 2024 were selected to create a report on the latest trends in the application of XAI techniques in intrusion detection systems.
- 4) Academic search engines, many websites, such as Medium and GitHub, were reviewed, especially for current XAI techniques and toolboxes.
- 5) Publications written in English were only included in this study.
- 6) The citations of the reviewed studies were also tracked to ensure comprehensive coverage of relevant research.
- 7) The results were carefully filtered by the authors to highlight the most reputable studies. The titles and abstracts of the filtered studies were scanned and selected according to the selection criteria, and the chosen studies were thoroughly reviewed.

1.5. Structure of this survey

The study presents background information about XAI, and then studies on IDSs and the use of XAI in IDSs are included. Section 2 provides comprehensive information about the definition, classification, explanation types, and goals of explainable artificial intelligence. Section 3 presents information about the importance, difficulties and studies in the literature of using XAI in IDSs. Sections 4 and 5 discuss the issues, challenges, open issues, and discussions for using XAI in IDSs.

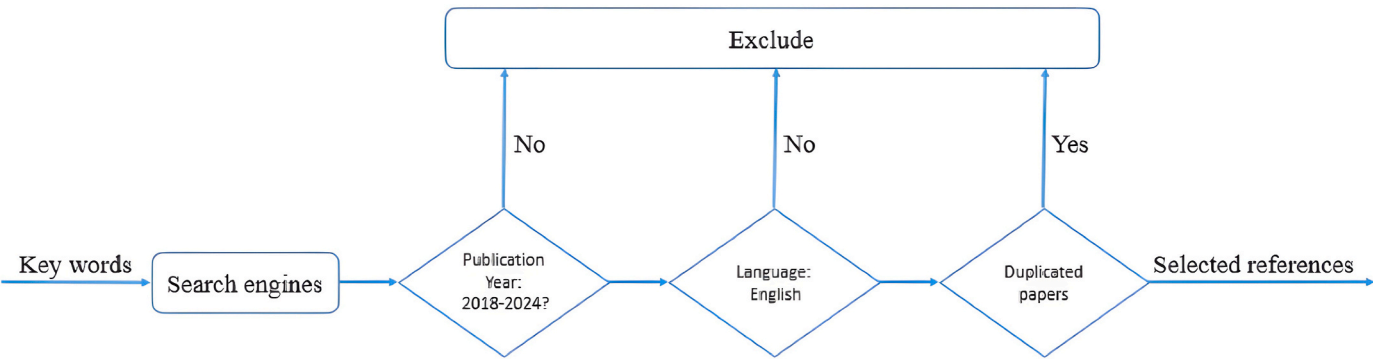


Fig. 3. Research methodology.

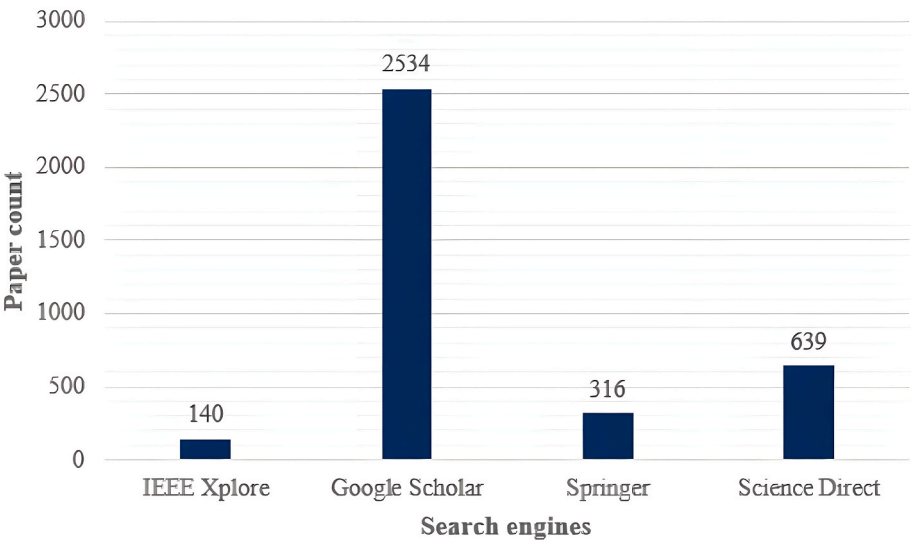


Fig. 4. Number of papers obtained according to search engines.

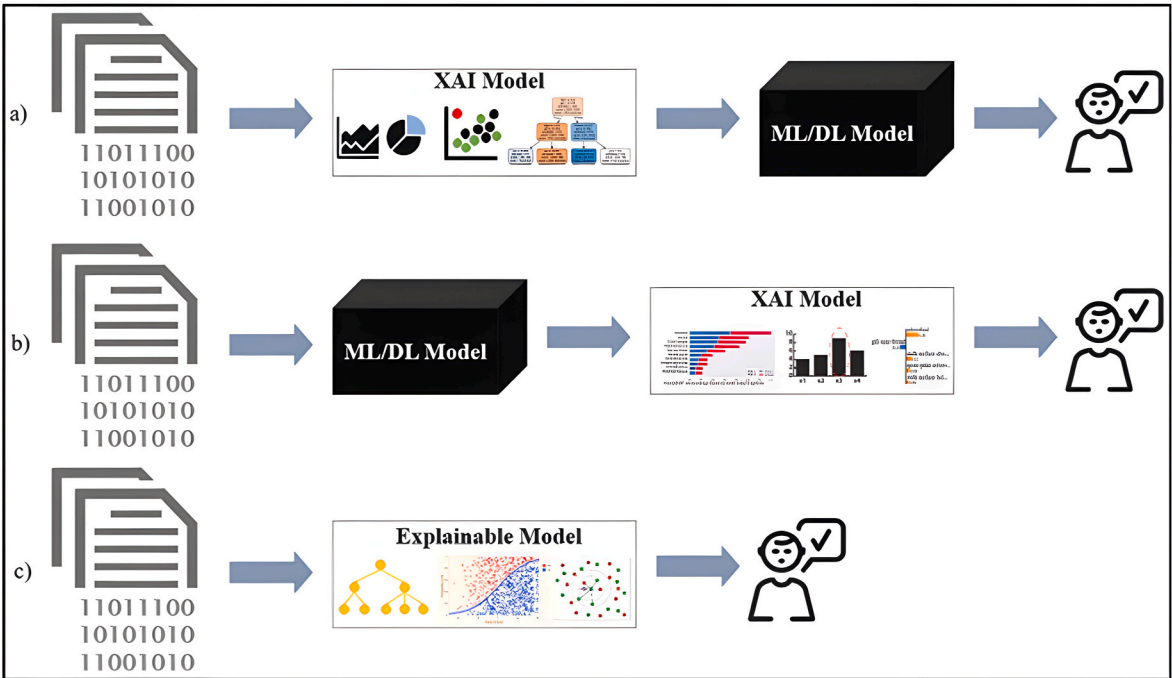


Fig. 5. XAI methods in terms of stages: a) Ante-hoc, b) Post-hoc, c) Intrinsic.

Finally, the conclusions and future directions obtained in Section 6 are mentioned.

2. Background knowledge of XAI approaches

XAI methods can provide explanations according to the stages of implementation of a model before, during or after the prediction as shown in Fig. 5. Ante-hoc methods involve external explanations before the training phase. Intrinsic methods refer to self-explanatory models that benefit from the internal structure of the model to provide explainability. Post-hoc methods refer to techniques that can be applied to any trained black box model without the need to understand the internal structure of the models. That is, it is not specific to any model but generates explanations for the predictions of all types of AI models.

As can be seen from Fig. 5, applications of AI and ML have become an crucial role in being a based model for the explainability. These models increasingly popular in various research and are used in many fields, such as cyber-attack detection, image classification, video recommendation, social network analysis, text mining, and natural language processing. Among various AI and DL algorithms as shown in Fig. 6, also known as representation learning (Deng, 2014), are widely used in these applications. The increasing growth and availability of data and significant developments in hardware technologies have led to the emergence of new research in the field of DL algorithms. DL approaches rooted in traditional neural networks significantly outperform previous AI algorithms. Many of the latest DL approaches have shown promising results in natural language processing (NLP), visual data processing, speech and audio processing, cybersecurity, and many other types of applications (Yilin et al., 2015, 2017).

An XAI technique is a recent approach that produces details, consequences, or reasons to make its functioning clear or easy to understand, how a decision is made, or get rid of from blackbox modelling limitations of AI models. XAI approaches mainly address the problem of interpretability and explainability of AI models.

As algorithms become more powerful and make better predictions with better accuracy, as shown in Fig. 6, it becomes increasingly important to understand how and why a prediction is made. The development of ML and DL algorithms has led to significant progress in improving the prediction and accuracy performance of various

applications, but it has also resulted in increasingly complex structures. The result of this situation is a trade-off between the development level and transparency of the algorithms. Without interpretability and explainability, predictions of real-life applications of AI are unlikely to be trusted. Human-understandable explanations increase system security as well as encourage trust and continued adoption of ML systems. As an emerging field, XAI is recently considered to be vital for critical decisions made by researchers and practitioners. Section 2 presents comprehensively the fundamentals, terminology, and taxonomy of XAI approaches.

2.1. XAI terminology

There is no concrete mathematical representation/definition for interpretability or explainability in general. There is also no clear metrics for explainability. However, a number of attempts have been made to clarify not only these two terms but also related concepts such as intelligibility (Lipton, 2018; Doshi-Velez and Kim, 2017; Gilpin et al., 2018). However, all these definitions lack mathematical formalism (Adadi and Berrada, 2018).

Most popular definitions of interpretability are that of Doshi-Velez and Kim's definition (Doshi-Velez and Kim, 2017) as "the ability to explain or present in understandable terms to a human being" and Miller's definition (Miller, 2019) as "the degree to which a person understands the reason for a decision", although these definitions are intuitive, these lack mathematical formalism (Adadi and Berrada, 2018). Explanations are formally defined as "an active feature of a model that refers to any action or procedure performed by a model for the purpose of classifying or detailing its internal functions" (Arrieta et al., 2020; Tjoa and Guan, 2021b). In addition, explanations refer to a series of meaningful outputs produced by the system for human understanding that can provide further information. These explanations explain why a particular decision was made. An example would be if a cyber defense system could explain why a particular behavior was labeled as attack. Explainability relates to the internal logic and mechanics within a ML system. The more explainable a model can be, the deeper the understanding people gain in terms of the internal procedures that occur when the model is trained or decisions are made (Linardatos et al., 2021).

Interpretability means that one can reveal what features contribute

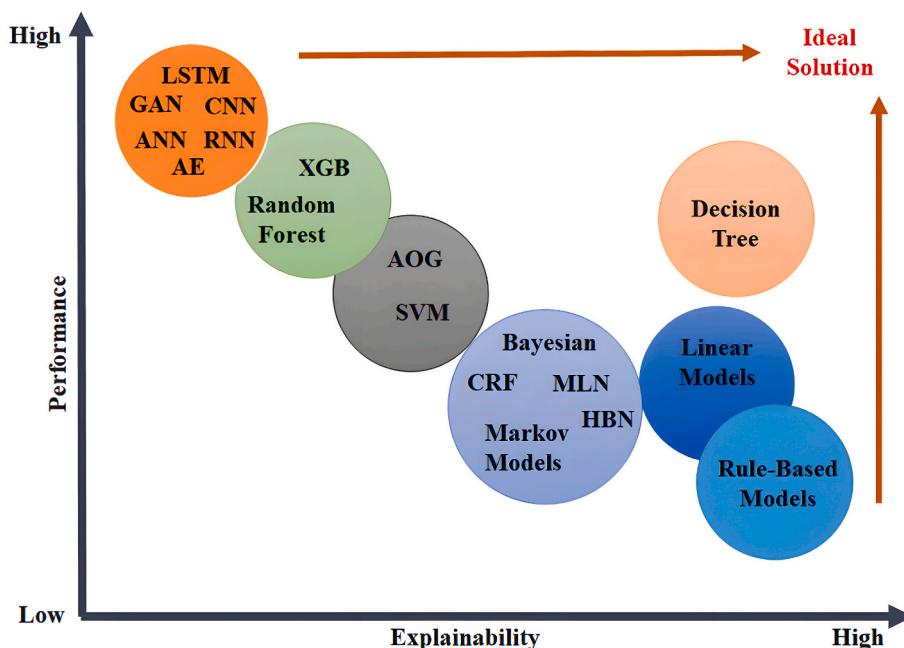


Fig. 6. The relationship between explainability and performance.

to a particular output and how the system made that decision by simulating the system. In general, interpretation focuses on the features that affect decisions, and helps to understand what kind of data is needed for the system to make certain decisions. Interpretability is often linked to the intuition behind a model's outputs. It can be considered that the more interpretable a ML system is, the easier it is to identify cause-and-effect relationships within the system's inputs and outputs (Adadi and Berrada, 2018).

2.2. XAI targets

Although AI provides significant benefits with its conveniences, it also involves several important problems, such as ethics, privacy, reliability, bias, opacity, and equality. XAI techniques can reduce these problems by addressing them in many ways. AI applications can be used and accepted safely by wider audiences and application areas in the future only if XAI methods achieve the goals specified in Fig. 7.

Informativeness is one of the main goals of XAI is to inform XAI users, researchers or experts. An explainability technique should be able to provide information about how, when, and under what conditions the AI model reached its conclusions.

Trustworthiness is an XAI algorithm should increase trust in AI algorithms with clear, easy-to-understand and accurately defined explanations. The way to achieve this, as stated in National Institute of Standards and Technology (NIST) (Phillips et al., 2021), is achieved by the trustworthiness of an AI application through its explainability. Trustworthy explanations give rise to trustworthy AI. Therefore, an XAI algorithm should aim to be trustworthy. Increasing the level of trust in XAI will naturally lead to better adoption by people.

In the last few years, the concept of **fairness** in AI has come to the fore with significant research on both data bias (Begley et al., 2020; Ntoutsis et al., 2020) and algorithmic bias (Garcia-Gathright et al., 2018). Explainability should allow us to determine if and when bias exists in AI models.

Causality is one of the main limitations of AI today is the lack of causality inherent in modelling (Kamath and Liu, 2021). In the field of XAI, more and more work is being done to identify and explain relationships in data (Janzing et al., 2020; Holzinger et al., 2019). Although significant domain and background knowledge is often

required to prove causality, explainability can be used to investigate cause and effect (Kamath and Liu, 2021).

Generalizability means that an XAI model trained on one task can be applied to different models and used as a starting point for other tasks. XAI also allows us to identify and understand what limits and limitations may exist in the generalization process of a model (Rai, 2019).

Stability is an AI model is expected to make the same decision under the same conditions. XAI should be able to give us insight into how stable a model will perform under various conditions.

The **accessibility** of AI applications by people without technical knowledge plays an important role in increasing popularity and adoption. XAI can facilitate the knowledge and understanding of complex AI models, thereby reducing the burden on ordinary people when dealing with them (Wolf and Ringland, 2020).

While the importance of **privacy** and security increases with AI applications, one of the advantages of XAI is the ability to evaluate the level of privacy (Kamath and Liu, 2021). With model explainability, it can be more easily evaluated whether confidentiality has been violated in encrypted data or algorithms (Vigano and Magazzeni, 2020). Additionally, XAI should be able to generate explanations without knowledge of the data, ensuring the integrity of the models and algorithms.

2.3. Types of explanations

XAI methods can be comprehensively presented with the six types of explanations as shown in Fig. 8.

The most common question encountered in AI systems is: "How a model works?" is the question. **Global explanations** attempt to answer this question and explain how models arrive at their predictions. Global explanations can be in the form of visual charts, mathematical formulas or model graphics. Global explanations are holistic explanations in that they provide us with a representation of the model's behavior.

When the question of how is answered with global explanations, the need to ask the question "why" arises. **Local explanations** seek to answer the question of why a model reaches a prediction for a particular input. It can match specific data or model algorithm features with their corresponding predictions.

Contrastive Explanations types of explanations help us understand



Fig. 7. Targets of XAI techniques.



Fig. 8. Types of explanation methods.

why, for a given input, a model produces the current prediction outcome rather than another prediction (Jacovi et al., 2021). They answer the questions “Why-not” or “Why not X and Y?” They are often used with “Why” explanations to understand a model’s prediction and expected behavior. They are useful in determining what minimal changes to inputs or model parameters are required to cause the model to make a different prediction.

What-if Explanations refer to hypothetical explanations of the changes in the model output when we change the inputs or model parameters of an AI model. It is a very useful type of explanation to help us understand the relationships between model predictions and model properties.

Counterfactual Explanations describe hypothetical changes in the inputs or parameters of a model that would cause the model to produce a desired output (Wachter et al., 2017). They answer the question of how to achieve the desired result by defining the most minor changes that can be made in the model without the need to understand the internal structure of the model (Byrne, 2019).

Example-Based Explanations are annotations that express the behavior of a model or underlying data distribution by simply highlighting specific instances of the data (Kim et al., 2016). A common usage choice is to provide examples of similar inputs from which the model will predict similar outputs.

2.4. XAI taxonomy

XAI taxonomy can be divided into different classifications according to different perspectives (Arrieta et al., 2020; Carvalho et al., 2019). It should be taken into consideration that classifications may change in parallel with the developing field of XAI. However, when XAI techniques are evaluated from different perspectives, a comprehensive XAI taxonomy is presented in Fig. 9. An XAI technique can be classified under one or more categories. For example, a method can be classified as post-hoc, model-specific, or global. Therefore, it is considered that it is a more accurate method to examine each technique under its own subclassification. This section presents a classification of the XAI family of methods according to their various approaches and features.

XAI methods can be global or local in terms of scope. While global

methods provide a holistic explanation of the behavior of models, local methods are used to explain behavior at the instance level. Global methods attempt to comprehensively explain the predictions of the entire model and provide an understanding of what effect model structures and parameters have on model outputs. This method provides an understanding of the entire model at once by providing an understanding of how the model’s input data are mapped to features and outputs (Kamath and Liu, 2021). Local methods, on the other hand, try to explain how a particular sample is matched with the output data. These methods explain how to reach the output of the model through an example. These methods reduce the structure to a small area of interest using a simpler model.

XAI can be grouped into model-specific and model-independent explanations based on implementation to specific classes of models. Model-agnostic methods are methods that can be applied to all types of models without limiting model classes. These methods distinguish between explanations and model classes. Thus, explanations become independent of model type. Most pre-model and post-model explainability methods are model-agnostic methods because they are applicable to many types of models.

Model-specific methods are methods created specifically for well-known examples of typical black boxes with complex and opaque structures, such as deep neural networks. As its name signifies, these methods only apply to the models created. Model-specific methods may be more advantageous than model-independent methods because they take advantage of certain features or architecture of the model. The disadvantage of model-specific explanations is that there is a limitation in model specification when a particular type of explanation is needed.

Feature importance representations in tree-based models, attention mechanisms in neural networks, and saliency maps are among the most well-known model-specific techniques. In tree-based models, feature importance approaches provide explanations by calculating feature importance scores that indicate which features have the greatest impact on the model’s predictions in AI models such as decision trees, random forests, and gradient boosting machines. These scores are derived from the structure of the trees and offer important insights into how the model makes its decisions. Attention mechanisms in neural networks are used in deep learning models to identify which parts of the input data the

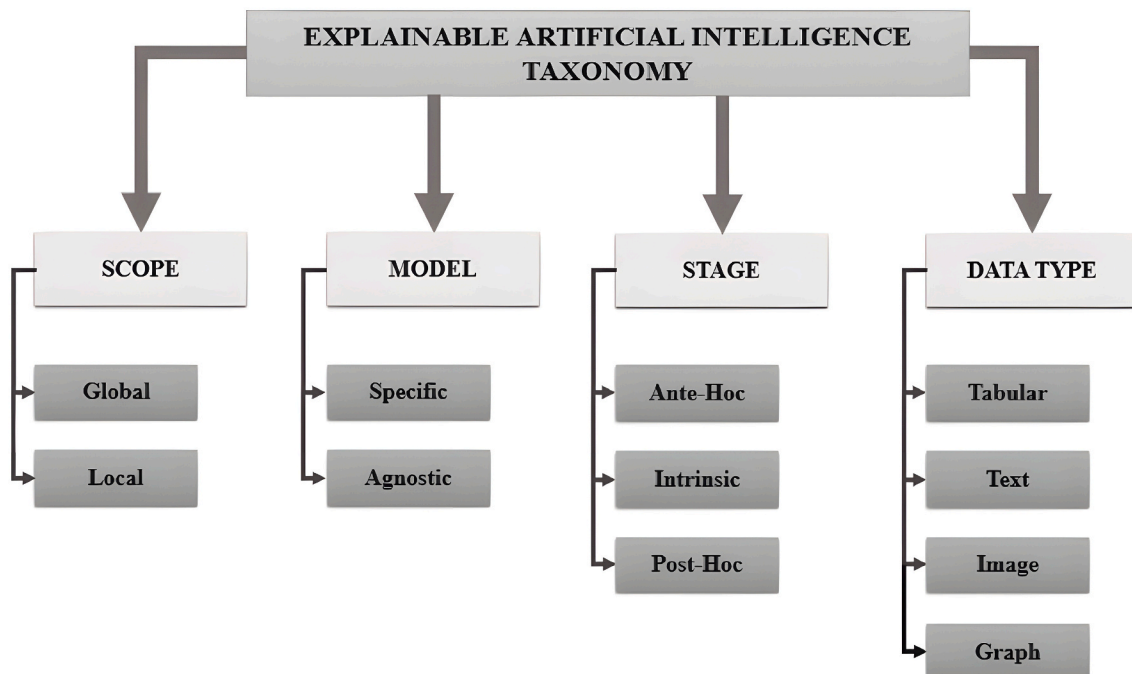


Fig. 9. XAI taxonomy.

model focuses on when making a prediction. For example, in a neural network used in an IDS, the attention mechanism can highlight specific packets or time intervals that are crucial for detecting an attack. Saliency maps are used to visualize which parts of an input image are most important for the prediction of an AI model using convolutional neural networks (CNNs). Although these approaches are primarily used in computer vision applications, similar techniques can be adapted for other types of data.

XAI can be divided into tabular, text, image, and graphic classes according to the output format of the created explanations.

The most well-known and frequently used techniques in the literature are LIME (Ribeiro et al., 2016), SHAP (Lundberg and Lee, 2017), and PDP (Friedman, 2001). LIME is a technique that explains individual predictions by locally approximating a black-box model with a simpler, interpretable model such as linear regression or a decision tree. LIME generates perturbed variations of the input data and observes how the model's predictions change. By analyzing these changes, LIME identifies which features in the input data have the most impact on the model's decision. SHAP provides a framework that combines game theory with local explanations to interpret predictions. It assigns a "SHAP value" to each feature, representing that feature's contribution to the model's prediction. SHAP values are based on the concept of Shapley values from cooperative game theory. SHAP can be applied to any AI model thanks to its model-agnostic structure. PDPs show the relationship between a feature or a set of features and the predicted outcome while keeping all other features constant. This technique visually explains the effect of a feature on the model's predictions.

2.5. XAI toolboxes

A single XAI method often does not fully do the task of disambiguating all relevant model features for the individual to be disclosed. Therefore, such toolboxes that implement multiple explainability methods in a single library with a common interface have emerged to solve this problem. In this section, information about some toolboxes published since 2018 is presented, and summarized in Table 2. The details of these toolboxes can be found in the references and the

available web links.

- **Skater**: Skater provides model explainability for all ML model formats and supports a total of seven XAI methods for the global and local explanation of different types of trained models, including DNNs for visual and textual inputs.
- **InterpretML**: This is an open-source Python library that provides explanations for both white-box algorithms like linear regression, rule-based models, decision trees, etc., and black-box algorithms via techniques like PDP, LIME, SHAP, etc.
- **iNNvestigate**: This library helps to provide explainability for various neural network architectures.
- **AI Explainability 360 (AIX 360)**: Developed by IBM, AI Explainability 360 includes eight state-of-the-art explainability methods and two evaluation metrics. **explainer**: Implemented as an add-on to the existing TensorBoard toolkit for the TensorFlow deep learning framework, explainer supports many post-hoc DNN explanations and analytics tools compatible with model understanding, diagnosis, and optimization pipeline stages.
- **FAT Forensics**: Based on Scikit-learn, this library includes various applications for model and data quality checks, as well as local and global explainability.
- **Alibi**: This is an open-source Python library that includes nine different XAI methods for classification and regression.
- **PDPbox**: It visualizes the impact of specific features on model predictions for supervised machine learning algorithms using partial dependency plots. Although PDPbox was initially designed for compatibility with all Scikit-learn models, different models may have different prediction interfaces.
- **ICEbox**: It implements Individual Conditional Expectation (ICE) plots, a tool for visualizing the predictions made by any supervised learning algorithm.
- **Captum**: An open-source model explainability library for PyTorch. Its three main features are multimodality, extensibility, and ease of use. The library also includes a visualization tool called Captum Insights, which helps with extraction, making it easier to use.

Table 2
XAI toolboxes.

Name of Toolbox	Reference	Input type	Local/Global	Specific/Agnostic	Code Repository
Skater	Choudhary (2018)	Tabular Image	Global	Agnostic	https://github.com/oracle/Skater
InterpretML	Nori et al. (2019)	Tabular	Local Global	Agnostic	https://github.com/interpretml/interpret
iNNvestigate	Alber (2019)	Tabular Image	Local Global	Agnostic	https://github.com/albermax/innvestigate
XAI 360 (AIX 360)	Arya et al. (2019)	Tabular Image	Local Global	Agnostic	https://github.com/Trusted-AI/AIF360
explainer	Spinner et al. (2020)	Image	Global	Agnostic	https://github.com/dbvis-ukon/explainer
FAT Forensic	Sokol et al. (2020)	Image Tabular	Local	Agnostic	https://github.com/fat-forensics/fat-forensics
Alibi	Klaise et al. (2021)	Tabular Image	Local Global	Agnostic	https://github.com/SeldonIO/alibi
PDPbox	(PDPbox)	Tabular	Global	Agnostic	https://github.com/SauceCat/PDPbox
ICEbox	Goldstein et al. (2015)	Tabular	Local	Agnostic	https://cran.r-project.org/web/packages/ICEbox/index.html
PyCEbox	(PyCEbox)	Tabular	Local	Agnostic	https://github.com/AustinRochford/PyCEbox
Captum	Narine et al. (2020)	Tabular Image	Global Local	Agnostic	https://github.com/pytorch/captum
H2O Explainability	(H2O documentation)	Tabular Statistical	Global	Agnostic	https://docs.h2o.ai/h2o/latest-stable/h2o-docs/explain.html
Ethical ML	(Ethical ML)	Tabular	Local Global	Agnostic	https://ethicalml.github.io/xai/index.html
Tf-Explain	(tf-explain)	Image	Local	Agnostic	https://tf-explain.readthedocs.io/en/latest/
What-If Tool	Wexler et al. (2019)	Tabular Image	Local Global	Agnostic	https://pair-code.github.io/what-if-tool/
ELI5	(ELI5 Documentation)	Tabular Image	Local Global	Specific	https://eli5.readthedocs.io/en/latest/
DALEX	Biecek (2018)	Tabular Statistical	Global	Agnostic	https://github.com/ModelOriented/DALEX

- **H2O**: A convenient wrapper for various explainability methods and visualizations in H2O. The main functions, `h2o.explain()` (global explanation) and `h2o.explain_row()` (local explanation), work for individual H2O models and some other models.
- **Ethical ML**: It includes a variety of tools that allow data analysis and model evaluation by analyzing the inferences of the results and the input features.
- **Tf-Explain**: It provides explainability methods for TensorFlow 2.0 to facilitate the understanding of neural networks. Feedback is provided on training models through both the core API and `tf.keras` callbacks.
- **What-If Tool**: It provides an easy-to-use, visual interface for explaining regression ML models and requires minimal coding and is integrated with Jupyter Notebook, JupyterHub, Colab, and TensorBoard.
- **ELI5**: It is an XAI approach that provides interpretable and explainable insights into the factors most relevant and influential in the model's predictions, using simple and intuitive language that non-experts can understand.
- **DALEX**: moDel Agnostic Language for Exploration and Explanation (DALEX) is a versatile R package that can be used to visualize and explain machine learning models.

XAI toolboxes offer specific advantages and limitations depending on different application scenarios and requirements. For example, Skater provides a model-agnostic approach, while InterpretML offers a more systematic framework with more comprehensive methods. Skater and explAiner stand out with their user-friendly interfaces, whereas tools like FAT Forensics and AIX 360 may require more expertise. Alibi, for instance, may necessitate a certain level of technical knowledge for users to interpret the explanations. Therefore, factors such as project requirements and model type should be considered when selecting toolboxes. In the future, further advancements in XAI are expected through the integration of these tools and the improvement of user experiences.

2.6. Explanation evaluation criteria

As presented in the previous sections, many methods of explanation are available, but there are still some questions that need to be answered in order to have a suitable explanation framework. How can we understand if the explanation is good? The question is one of the important questions. If it is stated that the explanation is good, how can we trust it is another important question. Additionally, any ML/DL model can be explained using any of the explainability methods, so there is uncertainty as to which XAI method to use or which XAI method is most appropriate for the underlying model and data used. To answer these questions, criteria that can evaluate explainability results are needed. In this section, explanation evaluation criteria were presented.

Various explainability evaluation taxonomies have been developed in the literature. In terms of logic and complexity, Doshi-Velez and Kim (2017) divided explanatory evaluations into three groups: application-based evaluation, human-based evaluation, and functional-based evaluation. Hoffman et al. (2019) mentioned common measurement methods in four aspects: goodness, satisfaction, understanding and efficiency measurement. Mohseni et al. (2021) have studied the evaluation of explanations through targeted audiences (e.g., AI experts) and metrics to verify the "efficiency" of explanations. Honnegger et al. (Honnegger, 2018) proposed three criteria to evaluate explainability. These criteria are the Identity criterion, which states that identical examples must have the same explanations; the Separability criterion, which states that the explanations of non-identical examples cannot be the same; and the Stability criterion, which states that similar examples must have similar explanations. Finally, the Defense Advanced Research Projects Agency (DARPA)'s XAI program evaluated XAI systems using the criteria of User satisfaction, Cognitive model, Task performance, and Trust evaluation (Gunning and Aha, 2019). In this study,

we categorized the criteria that will help evaluate explanations into two basic methods: computational criteria and cognitive criteria. However, evaluation methods such as User satisfaction, Cognitive model, Task performance, and Trust evaluation specified by DARPA have been considered sub-folds of cognitive measures. Similarly, evaluation methods such as Fidelity, Stability, Sparsity, Comprehensibility, Faithfulness, Local Accuracy, Robustness, Consistency, and Causality have been considered subcategories of computational metrics.

2.6.1. Computational metrics

These metrics indicate the numerical expression of the quality of the explanations produced in the XAI system (Mohseni et al., 2021). Measurements are usually made using appropriately developed numerical equations. Since it is based on numerical calculations, computational metrics can be used as a guide to create explanation techniques without human factors (Mueller et al., 2019). These metrics create computational ways to evaluate the accuracy of advanced explanations rather than using the study of human subjects. These metrics are widely used in the AI community because they describe the reliability of models with standard performance measures.

Fidelity (Lundberg and Lee, 2017) measures how closely the explanation matches the model's true predictions. In other words, it evaluates how similar the model's predicted outcome is to the explanation's prediction. The goal is to minimize the difference between the model's predictions and the explanation's output. Fidelity is formulated as follows:

$$Fidelity = 1 - \frac{\sum_{i=1}^n |\hat{y}_i - y_i|}{n} \quad (1)$$

where, $\hat{y}_i$ is the original prediction of the model, y_i is the prediction made through the explanation, and n is the total number of data points.

The Stability (Alvarez-Melis and Jaakkola, 2018) metric measures how consistent the explanations are for similar data points. If two similar data points have very different explanations, stability is considered low. Stability can be calculated as follows:

$$Stability = 1 - \frac{|Explanation(x_1) - Explanation(x_2)|}{d(x_1, x_2)} \quad (2)$$

where, $Explanation(x_1)$ and $Explanation(x_2)$ represent the explanations for two similar data points, and $d(x_1, x_2)$ defines the distance between the two data points.

Sparsity (Ribeiro et al., 2016) measures how simple an explanation is. Explanations that use fewer features are generally considered more understandable.

$$Sparsity = 1 - \frac{k}{m} \quad (3)$$

where, k is the number of features used in the explanation, and m is the total number of features used in the model.

Comprehensibility (Doshi-Velez and Kim, 2017) measures how easily the explanation can be understood by humans. This is usually related to the length of the explanation and the simplicity of the terms used. Instead of a mathematical formula for this metric, text analysis methods can be used. For example, understandability can be measured by calculating the word count and complexity of the explanation. Shorter and simpler explanations are generally considered more understandable.

Faithfulness (Lipton, 2018) measures how accurately the explanation reflects the actual workings of the model. If the explanation aligns with the model's decision-making mechanisms, faithfulness is considered high. Although the terms faithfulness and fidelity are sometimes used interchangeably, they serve different purposes and are often evaluated differently. At first glance, they may seem similar, but the distinction lies in how well the explanation is consistent with the model's outputs (fidelity) and how accurately the explanation reflects

the internal decision-making processes of the model (faithfulness).

2.6.2. Cognitive metrics

Cognitive measures are used to evaluate explanations that can be easily understood even by non-XAI experts. Since the primary goal of XAI is to make the logic and process behind the machine understandable to humans, evaluating explanations on human understanding is a clear evaluation indicator for explanations. Although computational metrics need to be taken with caution in specific tasks or applications, the evaluation of issues can be generalized to many situations by explicitly asking participants to identify the best explanation from their perspective (Hanif et al., 2021).

User satisfaction measures the accuracy and usefulness of the description based on the end user's opinions (Gunning and Aha, 2019). Some approaches, both subjective and objective, have been investigated to measure user satisfaction. Common approaches found in the literature are user interviews, Likert scale surveys, and expert case studies. Gedikli et al. (2014) used user interviews to investigate their satisfaction and the most effective ways to provide clarification. Studies in (Coppers et al., 2018) and (Lage et al., 2019) used a Likert scale survey to measure user satisfaction.

Cognitive model refers to a concept taken from cognitive psychology that explains how individuals think about how things work (Eldrandaly et al., 2023). A cognitive model is a depiction of how users understand the system (Mohseni et al., 2021). Cognitive models are derived from philosophical, psychological and natural models of human explanatory reasoning to measure the effectiveness of an explanation with its ability to predict the user's understanding of the system and their decisions in different situations (Gunning and Aha, 2019). These models provide insight into whether users should trust or doubt AI system decisions by assessing how much they understand the system and how they arrive at a particular decision. These measurements focus on understanding individual decisions, the overall model, the strengths and weaknesses of the system, and how its predictions are generated. Various studies have been conducted in the literature on these models (Lim and Dey, 2009).

Task performance measures whether explanation improves the user's decision-making process, as well as how well the user understands XAI systems. User task performance is the evaluation of the user's performance through completion time and success rate for a designated task supported by the system (Gunning and Aha, 2019). Various studies have been conducted in the literature regarding this metric (Lim et al., 2009; Lakkaraju et al., 2016).

Trust evaluation gets the user's ability to know when to trust or doubt decisions made by XAI systems (Gunning and Aha, 2019). Trust assessment is extremely important in an XAI system. Trust in an intelligent system is an important factor that affects whether the system is understood positively or negatively. This also represents a measure of the effectiveness of the system. User trust can be assessed by directly receiving user feedback through surveys and interviews during and after using a system. Various studies have been conducted in the literature regarding this metric (Yin et al., 2019; Nourani et al., 2019).

2.7. XAI in large language models (LLMs)

This section introduces the relationship between XAI and LLMs, which have become increasingly popular recently. Some evaluations have been made regarding the effects of these two important and developing fields of study on each other. This subject is included in this study because it is one of the important developments that will affect XAI's development and direction of change in the coming period.

LLMs refer to models that include DL algorithms used to understand and analyze natural language. LLMs can generate human-like text by predicting the probability distribution of the next word or sequence of words given the context of previous words. These models are often trained on large amounts of text data using DL algorithms such as recurrent neural networks (RNNs) or transformers. LLMs are unique in

their ability to perform various language-related functions, such as language translation, sentiment analysis, chatbot dialogues, and more. LLMs are extremely popular due to their potential to perform various natural language processing activities such as text summarization, language translation, question answering, and text completion. OpenAI's GPT (Generative Pre-trained Transformer) series is one of the most well-known LLMs.

LLMs have achieved revolutionary success in how machines understand and produce human-like text. However, aside from their extraordinary capabilities, the internal mechanisms of these models are still unclear, and this lack of transparency poses undesirable risks for applications. Therefore, understanding and explaining these models is of great importance in terms of elucidating their behavior, limitations and social impacts (Haiyan et al., 2024). LLMs have some challenges in ethics and interpretability in terms of privacy, hallucination, and trust. Because LLMs process large amounts of data, privacy concerns have emerged as a significant challenge. It is imperative to ensure that personal and sensitive information is handled responsibly and to prevent unintended consequences. Ethical approaches that strike a balance between the performance of models and the protection of user privacy need to be established. Hallucination refers to the production of false or fabricated information by LLMs. These models can cause misinformation by producing false or misleading content with a hallucination effect. Finally, users must trust that the content created is reliable and unbiased. Therefore, ensuring transparency in the decision-making processes of LLMs and providing understandable explanations for the outputs produced is important in establishing trust.

The increasing popularity of LLM models in recent years has brought about the problem of trust in these models. Therefore, the focus of XAI has recently shifted towards LLMs, which are often criticized for their lack of transparency and explainability (Wu et al., 2024). Meeting the explainability need of LLMs will provide significant development and transformation in both the XAI and LLM fields. Due to their complex structure, many of the existing XAI methods cannot be directly applied to LLMs, making developing new and domain-specific explainability approaches necessary. In addition, the trust and transparency capabilities that XAI will provide to LLMs will continue to actively increase their productivity and applicability. As a result, XAI and LLM, two popular fields in recent years, will continue to mutually develop and change each other. Various explainability approaches have been developed in the literature for the explainability of LLMs such as The Learning Interpretability Tool (Tenney et al., 2020), Phoenix (Lin et al., 2022), and Comgra (Murari et al., 2022). In addition to addressing the explainability needs of LLMs with XAI, it is thought that using LLMs for XAI can provide significant contributions. The key contributions of using LLMs for XAI applications can be as follows (Zytek et al., 2024; Cambria et al., 2024; Cheng et al., 2024; Lubos et al., 2024; Castelnovo et al., 2024; Kunz and Kuhlmann, 2024; Khediri et al., 2024; Ajwani et al., 2024).

- While traditional explanations focus on the mathematical and technical aspects of the model, LLMs can explain the outputs of a complex model to wider audience with smoother and deeper explanations.
- LLMs can simplify the decision-making processes of a complex model, especially in DL model, but XAI model increase the complexity of the system making across millions of parameters and layers.
- LLMs can explain not only the result but also how the model arrived at that result, making it easier to produce explanations based on cause-and-effect relationships directly but XAI models are still static.
- While traditional explanations focus on input data, layers, connections and decisions, LLMs can provide more in-depth technical explanations for experts, simpler and more understandable explanations.

- While XAI models are dependent of models' parameters, data, mathematical expressions or rules, LLMs are interactive with users or prompts in depth of explanations.

The use of LLM techniques in XAI applications might also offer significant advantages but also brings with it certain challenges and limitations (Ajwani et al., 2024; Perković et al., 2024; Barman et al., 2024; Jiao et al., 2024; Ayyamperumal and Ge, 2024). These difficulties and limitations are.

- Risk of hallucination in producing proper outputs due to missing information in the model.
- Lack of LLM transparency in explaining certain parameters and producing proper explanations suitable for a particular usage scenario or context.

- Missing explanations of inner mechanisms of LLMs for risky or critical issues such as cybersecurity, medicine, law, etc.
- Many facilities and opportunities are available in the literature for flexible and adaptable explainability considering dynamic LLMs, XAI models are still static.

3. XAI in intrusion detection systems

Intrusion refers to any unauthorized activity that disrupts or damages an information system. That is, any attack that violates the integrity, confidentiality or availability of an information system is defined as an intrusion. The purpose of IDS is to detect any abnormal activity in an information system that cannot be detected by basic security devices such as firewalls. Therefore, an IDS has become a necessary tool in computer networks to provide a more secure network environment.

IDSs are classified according to many different criteria. The most

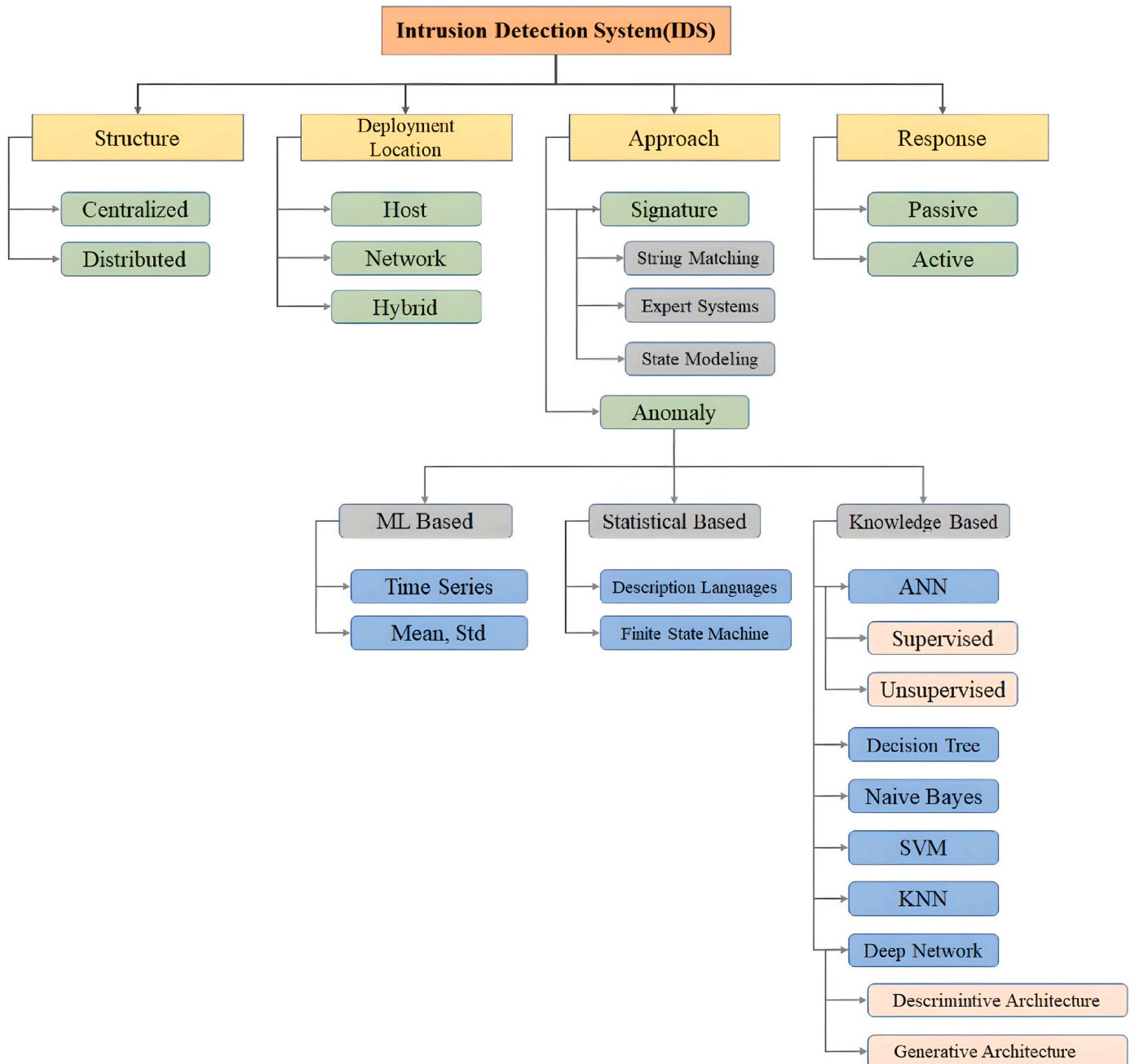


Fig. 10. IDS classification (borrowed from (Eshakagdy et al., 2022) and modified).

well-known type of classification is based on the attack detection method and is divided into two: anomaly detection and misuse detection. However, different classifications can be made, such as the architectural structure of IDSs, the type of the protected system, and the time of data processing, as shown in Fig. 10 (Eshakagdy et al., 2022). While the approach based on anomaly detection models user behavior in the system, the approach based on misuse (signature) detection models the behavior of attackers.

AI is increasingly being used in cybersecurity solutions. Traditional IDSs include rule-based or signature-based techniques. However, in recent years, AI-enabled IDSs have emerged as an effective security measure to combat attacks. These IDSs can detect abnormal activities by pre-learning most of the connections and traffic in a network infrastructure where traditional IDSs cannot perform adequately. Additionally, intelligent IDSs are required to counter new forms of attacks or cleverly planned intrusions that emerge every day.

AI techniques can even detect anomalies that are difficult to detect by a human. For example, if an attacker exhibits abnormal behavior in network traffic that cannot be noticed by experts, AI-based IDSs can notice this and detect attacks. The ability of AI techniques to detect even small anomalies distinguishes them from other types of traditional IDS (Nwakanma et al., 2023). As attacks evolve day by day and new vulnerabilities constantly emerge, it is invaluable for AI-based models to have the ability to adapt and grow. Manually defined rule-based approaches require significant effort to maintain and update the existing rule set (Šarčević et al., 2022). There is a trend toward using ML models to reduce manual work requirements and the number of experts required to create and analyze rules. This is considered to be the reason why

signature-based IDSs are becoming obsolete, thus giving way to anomaly-based intelligent IDSs as a new trend.

IDSs containing ML and DL methods have been applied in many studies and have been shown to achieve very good experimental results. Many of these studies use ML techniques to improve the classification accuracy of known attacks, recognize anomalous network traffic, and automate model building. However, the increasingly complex models with superior capabilities in these studies act like a black box (Kumar et al., 2010). However, few of these systems focus on explaining the results as a way to understand how ML algorithms arrive at their conclusion to predict attacks. Although these models have been successfully applied in most studies, no plausible explanation for the models has been provided. The ability to explain an AI algorithm's decisions provides many important benefits, including greater trust in the system and a better understanding of its operation. Therefore, the need for explainability of DL-based IDSs, which are increasingly used and have achieved significant success, is increasing.

3.1. Intrusion detection system datasets

In the field of cybersecurity, especially in IDSs, the judicious selection and use of datasets play an important role in developing, training, and evaluating various machine learning and statistical models (Yavanoglu and Aydos, 2017). Datasets are essential for validating any IDS approach by allowing us to evaluate the method's ability to detect behaviors. Similarly, the quality and capacity of the data significantly influence the decisions of XAI models. Although numerous software tools, such as Wireshark or WinDump, are used to generate

Table 3
Overview of IDS datasets.

Name of Dataset	Reference	Year	Instance Number	Features Number	Labeled Data	Normal Traffic	Attack Traffic	Attack Types
NSL-KDD	NSL-KDD Data Set ,	2009	150K	41	✓	✓	✓	Dos,R2L, U2R, Probe
UNSW-NB15	Moustafa and Slay, 2015	2015	2M	49	✓	✓	✓	DoS, worms, Backdoors, and Fuzzers
CIC-IDS2017	CICIDS2017 Dataset ,	2017	3.1M	80	✓	✓	✓	Brute force, Portscan, Botnet,Dos,DDoS, Web, Infiltration
ADFA-LD	Crech and Hu, 2013	2013	2.7M	26	✓	✓	✓	Zero-day attacks, Stealth attack, C100 Webshell attack
ToN_IoT	Alsaedi et al., 2020	2020	22.3M	132	✓	✓	✓	XSS, DDoS, DoS, password cracking attacks, reconnaissance or verification, MITM, ransomware, backdoors, and injection attacks
CIC-IoT-Dataset-2022	IoT Dataset ,	2022	1M	48	✓	✓	✓	Flood and RTSP- Brute Force
CSE-CIC-IDS2018	CSE-CIC-IDS2018 Dataset ,	2018	8.3M	80	✓	✓	✓	Brute force, Portscan, Botnet, Dos, DDoS, Web, Infiltration
CPS Dataset	Faramondi et al., 2021	2021	24.3M	14	✓	✓	✓	DoS, scanning and MITM attacks
CICDDoS2019	Sharafaldin et al., 2019	2019	50M	80	✓	✓	✓	12 different DDoS attacks
IoT-23	Garcia et al., 2020	2020	325M	19	✓	✓	✓	20 different malware attacks
MalDroid20	Malicious Android Dataset ,	2020	11K	470	✓	✓	✓	Adware, Banking and SMS malware, Riskware
KDD99	KDD Cup ,	1999	5M	41	✓	✓	✓	Dos, R2L, U2R, Probe
BoT-IoT	Koroniots et al., 2019	2019	1.5M	34	✓	✓	✓	DoS, DDoS, Port scanning, Brute force, Information theft, CC
UGR'16	Maciá-Fernández et al., 2018	2016	16M	12	✓	✓	✓	DoS, Scan, Botnet, SPAM
Kyoto	Song et al., 2011	2006	93M	24	✓	✓	✓	Shellcode and malware
ISCX-URL2016	Mamun et al., 2016	2016	114K	17	✓	✓	✓	Spam, Malware, Phishing, and Defacement.
ISCXIDS 2012	Shiravi et al., 2012	2012	2M	20	✓	✓	✓	Brute force, infiltrating, HttpDos, DDoS
ISCX VPN-nonVPN	Gil et al., 2016	2016	45K	63	✓	✓	✓	Chat, VoIP, Streaming service groups
AWID	Kolas et al., 2016	2016	37M	15	✓	✓	✓	authentication request, ARP flooding, injection, probe request
CIDDS-001	Ring et al., 2017	2017	32M	81	✓	✓	✓	DoS, port scans (ping-scan, SYN-Scan), SSH brute force
Botnet	Beigi et al., 2014	2014	19K	25	✓	✓	✓	Menti, Murlo, Neris, NSIS, Rbot, Sogou, Strom, Virut, Zeus
CDX	Sangster et al., 2009	2009	500K	75	✓	✓	✓	Buffer overflow

cybersecurity data, these methods are generally more suitable for collecting narrow or low-volume data. However, the datasets required for intrusion detection systems often involve high acquisition time and costs (Xin et al., 2018). Therefore, using publicly available cybersecurity datasets from the literature can reduce the time spent on data collection and increase study efficiency. Researchers can train, validate, and evaluate XAI-based cybersecurity solutions using these datasets. Detailed descriptions of several important datasets used in IDS research are presented in Table 3. Each dataset is analyzed in terms of its source, size, and composition.

In order to compare different intrusion detection datasets and help researchers find suitable datasets for their specific scenarios, features such as year, attack types, normal and attack traffic, number of instances, and number of features were used as the basis of evaluation. Since new attack scenarios emerge every day, the recency of an intrusion detection dataset plays an important role in its usability. The "year" feature was used to indicate the dataset's up-to-dateness. The presence of normal traffic data was used as another comparison parameter, with values of either "yes" or "no." The inclusion of normal user behavior is one of the essential parameters for evaluating an IDS. IDS datasets should contain various attack scenarios or types, indicating the type of malicious network traffic in the dataset. The "labeled data" parameter indicates whether the datasets are labeled or not. Labeled datasets are necessary for training supervised ML methods and evaluating intrusion detection techniques. In addition to these parameters, the number of instances and the number of features contained in the datasets were also used to guide researchers in their selection.

3.2. Importance of XAI in IDSs

High performance, generalizability, flexibility, and the ability to detect zero-day attacks are important benefits of AI techniques in the field of cybersecurity. In addition, AI-based approaches can process dynamic data that changes over time, recognize large-scale and multi-dimensional data, and identify anomalies in the data. However, DL-based approaches have many drawbacks, such as a lack of flow information, vulnerability to evasion attempts, limited data knowledge required to design relevant features, and lack of qualified domain experts to review the application (Capuano et al., 2022). Particularly, the last two drawbacks raise the need for explainability. Additionally, it is critical for AI models to have explainability and interpretability for cybersecurity experts, researchers, and developers to understand these models, their values, and the accuracy of their findings. Explanations are needed to investigate false alarms, identify system biases and errors, and make informed decisions for future system improvements (Šarčević et al., 2022).

While the increasingly developing and complex structures of the techniques used in intelligent IDS models increase the success of the models, they also reduce their transparency, making it difficult to adopt intelligent IDSs that turn into black boxes. Many IDS models have not had the chance to be used in practical applications because their results do not provide explanations that will enable cybersecurity experts to optimize their decisions. Therefore, cybersecurity experts, analysts, and decision-makers are reluctant to trust decisions made by intelligent IDSs based on AI models (Amarasinghe et al., 2018). One of the most important factors affecting confidence in any IDS detection is understanding the impact of malicious data on detecting any attack on the system. Most studies in the literature have focused on the accuracy of various classification models rather than trusting IDS. XAI techniques increase trust in IDSs by allowing cybersecurity experts to understand the data evidence and causal reasoning underlying the decisions of AI-based intelligent IDSs (Le et al., 2022).

In the event of a cybersecurity violation, it is often difficult for an expert to understand the logic of the AI system. Additionally, it is difficult to make decisions in such a black-box environment. AI systems sometimes provide false findings in the form of false positives and false

negatives, which mislead security experts and jeopardize the integrity of the entire system. In such cases, there is a serious need for explainability capabilities to more accurately handle violation scenarios or abnormal events. XAI provides security professionals with basic reasoning and threat detection capabilities to reduce the risk of violations and improve overall security. XAI also has capabilities to assist with risk discovery and prioritization, incident response coordination, and malware threat detection. XAI appears to be a promising solution in situations requiring explainability, interpretability, and accountability (Srivastava et al., 2022). An IDS needs to provide simple explanations for the questions of which part of the information system infrastructure the attacks target, which vulnerabilities of the information systems they exploit, or which security policies are violated. The main solution that can meet this need is IDSs, which have models that can logically explain the results. In summary, explanations in IDSs help users better understand what the pattern in an intelligent IDS is, what features of an instance are malicious and why it is classified as an attack, and how the IDS's judgments can be trusted.

3.3. XAI challenges in IDSs

Unlike other application areas, cybersecurity poses particular challenges regarding the need for explanation methods (Warnecke et al., 2020). Firstly, due to the importance of the results produced by cyber security systems, it is a matter of complete trust in these systems. Another major problem consists of a large number of learned model biases and human biases in security applications (Murdoch et al., 2019). Learned model biases are typically due to imbalanced data, historical inequities, or even systematic issues in the way data is collected and labeled. For example, a IDS trained on more instances of cyber attacks from certain geographic regions might unfairly prioritize or overlook risks from others. If an AI model reflects these biases, it can provide explanations that reinforce these inaccuracies. This undermines the trust in the AI's decision-making process because the explanations might suggest that certain security threats are more prevalent in specific regions, demographics, or behaviors without justifiable reason. XAI systems can struggle to explain away these biases, and they may inadvertently reinforce harmful or incorrect security conclusions. Human biases arise when security analysts or operators interpret or interact with AI systems. These can include cognitive biases, such as confirmation bias (favoring information that supports pre-existing beliefs), anchoring bias (relying too heavily on the first piece of information encountered), and other subjective influences. When humans are involved in reviewing or verifying the outputs of XAI systems, their biases can affect how they interpret the explanations. For instance, if a human operator already suspects a certain type of threat, they might selectively focus on the parts of the AI's explanation that confirm their suspicions, ignoring contradictory information. This can lead to skewed decision-making, where the human and machine biases amplify each other. Data-driven decision-making models can inherit these biases, and subsequent predictions from such models may be unfair or even inaccurate. Explanations in security applications can indicate underlying causalities in the model (Yan et al., 2022).

As with all new technologies, the implementation of XAI in IDSs brings with it some disadvantages as well as its advantages. XAI can provide significant improvement through the annotations it provides to IDS applications, but it can also facilitate new attacks on AI applications for the attacker, which could pose serious security threats since IDSs would be explainable (Taddeo et al., 2019). However, when the advantages and disadvantages are evaluated together, as in open source discussions, XAI is considered to have more advantages than disadvantages by reducing the risks to the adoption of AI. Additionally, defining AI models compatible with XAI principles or developing model-agnostic XAI frameworks will increase capabilities to recognize vulnerabilities in a timely manner, allowing large-scale AI use in industrial and human scenarios (Capuano et al., 2022).

Another challenge is the trade-off between accuracy and explainability. Simpler models, such as linear models and decision trees, are more interpretable than complex models, but they may not provide the desired accuracy. This is a significant challenge in IDSs, where both high accuracy and explainability are crucial.

Scalability and computational load can also pose significant challenges for the use of XAI in IDSs. Some XAI techniques, such as SHAP and LIME, may not be computationally efficient, especially when applied to large datasets or complex models. This can be a limitation in real-time IDS environments where immediate responses to threats are required.

The challenge of generating context-specific explanations arises as well when using XAI in IDSs. The effectiveness and efficiency of XAI techniques can vary depending on the context in which they are applied. A model that is explainable in one context may not provide meaningful explanations in another.

The usefulness of explanations can also vary depending on the expertise of the users. The level of detail and technical complexity in the explanations should be adapted to the target audience. This creates the challenge of generating explanations that match the expertise level of different users.

3.4. XAI Publications in IDS

In this section, XAI-enabled IDS studies in the literature are discussed in detail, and the information obtained from these studies is presented in Table 4. Additionally, the studies are examined from various perspectives, such as the XAI techniques used by the researchers, AI models, datasets, whether the XAI techniques are model-specific, and whether they are intrinsic or post-hoc. A summary of the existing studies from these perspectives is presented in Table 5.

3.4.1. XAI techniques in intrusion detection systems

SHAP is a widely adopted XAI method in IDS studies due to its ability to provide both global and local explanations for model decisions by assigning importance scores to individual features, as shown in Fig. 11. SHAP has been preferred in 82 different IDS studies, as indicated in Table 5. The greater use of SHAP in IDS studies compared to other models highlights its flexibility and strength in offering both local and global feature importance explanations. These features of SHAP have proven to be a powerful solution for increasing confidence in IDS decisions, especially when complex models such as deep neural networks (DNNs) are used.

LIME is another popular XAI method that explains individual predictions by approximating a complex model with a locally interpretable one. LIME's ability to explain individual predictions is particularly useful in cases where security experts need to understand why certain examples are classified as attacks, as shown in Fig. 12. This advanced capability has made it the most preferred method in the literature after SHAP.

In particular, self-explainable methods such as Random Forest and Decision Tree have also been used for explainability in IDS studies. As shown in Fig. 13, decision trees do not require post-hoc explainability methods like SHAP or LIME since they are inherently interpretable. However, for random forests, which combine decisions from multiple trees, post-hoc explainability methods such as SHAP can be used to explain the contribution of each feature in the ensemble. As seen in Table 4, there are many different XAI techniques used in IDS systems in the literature, beyond SHAP, LIME, and self-explainable methods.

Most of the studies in the literature have used previously developed XAI methods to make the results explainable, and as a result of this situation, the explanations in most of the studies are post-hoc. In the studies, SHAP is the most adopted method. After SHAP, LIME is another XAI method used in the studies. However, in some studies, various methods such as decision tree and random forest, which are white-box in nature and provide the necessary explanations, have been used.

3.4.2. AI models in intrusion detection systems

In this section, the current status of various AI models in XAI-supported IDS studies is examined, ranging from traditional ML models such as decision trees, random forests, and gradient boosting machines to DL models such as CNNs and RNNs. In the studies found in the literature, a wide variety of AI methods have been used, with RF, DT, DNN, XGBoost, ANN, and CNN techniques being the most adopted methods, respectively.

As shown in Table 5, decision tree and random forest algorithms have been frequently preferred in XAI-supported IDS studies. Decision trees and random forests are among the most interpretable AI models in machine learning, which is why they have been commonly used in XAI studies. Additionally, these algorithms are often employed alongside traditional ML algorithms such as SVM, MLP, NB, KNN, and LR to evaluate prediction performance in IDS studies.

Following traditional ML algorithms, deep learning-based techniques such as ANN, DNN, CNN, LSTM, GBM, and Autoencoder have been widely preferred in XAI-supported IDS systems, especially in recent years. These techniques are favored because they significantly outperform traditional ML techniques in terms of predictive performance. While DNN and ANN algorithms form the basis of DL-based techniques, specialized DL models like CNN and LSTM are also frequently used in the literature.

Gradient Boosting Machines (GBM), including popular variants such as XGBoost and LightGBM, are among the best-performing machine learning models for tabular data. These models are essentially based on decision trees, but since GBM models are inherently more complex than decision trees, they require post-hoc explanation methods such as SHAP or LIME.

CNNs are another AI technique frequently used in IDS. CNNs are inherently complex and difficult to interpret. While algorithms such as SHAP and LIME are often used to enhance the explainability of CNNs, CNN-specific methods such as Grad-CAM (Gradient Weighted Class Activation Mapping) and LRP (Layer-Based Relevance Propagation) are widely used to create visual explanations.

RNNs and their variants, such as Long Short-Term Memory (LSTM) networks, are designed to process sequential data. They are widely used in time series prediction, speech recognition, NLP and cybersecurity. Due to their sequential nature, RNN and LSTM models can be difficult to explain. Post-hoc methods such as LIME are often employed to explain the effect of individual time steps or words in a sequence. Additionally, attention mechanisms often integrated into RNNs provide a built-in form of interpretability by indicating which parts of the input sequence the model focuses on more heavily.

Autoencoders are unsupervised neural networks used for tasks such as dimensionality reduction, anomaly detection, and data compression. They work by encoding the input data into a lower-dimensional space and then reconstructing it back to its original form. In anomaly detection, autoencoders are used to identify outliers by measuring the reconstruction error, which is typically higher for anomalous data. SHAP and LIME can be used to explain why certain data points are flagged as anomalous, providing insights into the underlying patterns the model has learned.

3.4.3. Datasets in intrusion detection systems

As explained in detail in Section 3.1, there are many datasets in the literature that can be used in IDS studies. The most frequently used dataset in IDS studies is NSL-KDD. This dataset is preferred more because it is an older dataset, and its structure is suitable for many AI models. Following the NSL-KDD dataset, the UNSW-NB15 and CIC-IDS2017 datasets, which have become increasingly popular among researchers in recent years, are commonly used. In addition to these datasets, many others used for specialized tasks have been employed in the literature, as shown in Table 3. When the table is examined, it is evident that the KDD99 dataset is no longer preferred in many studies, despite being the oldest dataset. The reason for this is that it has become

Table 4
XAI articles in IDSs.

Reference	Year	ML/DL Models	Dataset	XAI Models	Local/ Global	Model Specific/ Agnostic	Post-hoc/ Intrinsic
Gaspar et al. (Gaspar et al., 2024)	2024	MLP	ADFA-LD	SHAP, LIME	Local, Global	Agnostic	Post-hoc
Hoosmand et el. (Hoosmand et al., 2024)	2024	XGBoost	NSL-KDD, UNSW-NB15	SHAP	Global	Agnostic	Post-hoc
Ahamed Maricar et al. (Ahamed Maricar et al., 2024)	2024	XGBoost	CIC-IDS2017	LIME	Local	Agnostic	Post-hoc
Mallampati and Hari (Mallampati and Hari, 2024)	2024	LGBM	CIC-IDS2017, UNSW-NB15	SHAP	Local, Global	Agnostic	Post-hoc
Sivamohan and Sridhar (Sivamohan and Sridhar, 2023)	2023	BiLSTM	NSL-KDD, HoneyPot	LIME, SHAP, KHO	Local, Global	Agnostic	Post-hoc
Marwa et al. (Marwa et al., 2023)	2023	LSTM	NSL-KDD, UNSW-NB15 and TON_IoT	SHAP, ICE, PDP, PFI	Local, Global	Agnostic	Post-hoc
Lanfer et al. (Lanfer et al., 2023)	2023	RF	NSL-KDD	SHAP	Global	Agnostic	Post-hoc
Siganos et al. (Siganos et al., 2023)	2023	–	CIC-IoT-Dataset-2022, IEC 69870-5-104	SHAP	Local, Global	Agnostic	Post-hoc
Manh-Dung et al. (Manh-Dung et al., 2023)	2023	CNN, SAE	CSE-CIC IDS2018	SHAP	Local, Global	Agnostic	Post-hoc
Sherif and Mohamed (Sherif and Mohamed, 2023)	2023	RF, SVM, NB, MLP, CNN	CPS	SHAP	Local	Agnostic	Post-hoc
Almuqren et al. (Almuqren et al., 2023)	2023	IENN-based classification	CIC-IDS2017, NSL-KDD	LIME	Local	Agnostic	Post-hoc
Sharma et al. (Sharma et al., 2023)	2023	DNN	NSL-KDD	LIME	Local	Agnostic	Post-hoc
Rehman et al. (Rehman et al., 2023)	2023	CNN	NSL-KDD	LIME	Local	Agnostic	Post-hoc
Youcef et al. (Youcef et al., 2023)	2023	LSTM	NSL-KDD	SHAP	Local	Agnostic	Post-hoc
Pande and Khamparia (Pande and Khamparia, 2023)	2023	DNN	NSL-KDD, UNSW-NB15	SHAP	Local, Global	Agnostic	Post-hoc
Sharma et al. (Sharma et al., 2024)	2023	DNN, CNN	NSL-KDD, UNSW-NB15	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Sevri and Karacan (Sevri and Karacan, 2023)	2023	DNN	–	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Larriva-Novo et al. (Larriva-Novo et al., 2023)	2023	DT, XGBoost, LR, KNN, RF, MLP	UNSW-NB15	SHAP	Local	Agnostic	Post-hoc
Gurbuz et al. (Gurbuz et al., 2023)	2023	KNN, DT, RF, NB, SVM, MLP, ANN	Intensive Care Unit(ICU)	LIME, SHAP, ELI5, IG	Local, Global	Agnostic	Post-hoc
Lavieille and Atlas (Lavieille and Atlas, 2023)	2023	Isolation Forest, GMM, SVM, AE, LOF	Microsoft Defender's logs	SHAP	Local, Global	Agnostic	Post-hoc
Chinaechetam et al. (Chinaechetam et al., 2023)	2023	LGBBoost	Edge-IIoTset	SHAP	Local	Agnostic	Post-hoc
Recio-García et al. (Recio-García et al., 2023)	2023	DT, AdaBoost, LR, NB, RF, MLP	SQL injection dataset	FI, PI, SHAP	Local	Agnostic	Post-hoc
Kalutharage et al. (Kalutharage et al., 2023)	2023	Autoencoder	USB-IDS	SHAP	Local, Global	Agnostic	Post-hoc
Javeed et al. (Javeed et al., 2024)	2023	BiLSTM, Bi-GRU	CICIDDoS2019	SHAP	Global	Agnostic	Post-hoc
Bacevicius and Paulauskaite-Taraseviciene (Bacevicius and Paulauskaite-Taraseviciene, 2023)	2023	LR, DT, RF, CNN, ANN	CIC-IDS2017 CSE-CIC-IDS2018	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Patil et al. (Patil et al., 2023)	2023	DT, SVM, KNN, NN	CPS dataset	SHAP	Local	Agnostic	Post-hoc
Muna et al. (Muna et al., 2023)	2023	XGBoost	IoTID20	ELI5, LIME, SHAP	Local, Global	Agnostic	Post-hoc
Hassan et al. (Hassan et al., 2023)	2023	RF	DoS dataset	LIME	Local	Agnostic	Post-hoc
Lanvin et al. (Lanvin et al., 2023)	2023	Autoencoder	CIC-IDS2017	Self explainable	Local, Global	Specific	Intrinsic
Pawlicki et al. (Pawlicki et al., 2023)	2023	DNN	DDoS Backscatter	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Zhou et al. (Zhou et al., 2023)	2023	DT	CIC-AWS-2018	Self Explainable	Global	Agnostic	Post-hoc
Arreche et al. (Arreche et al., 2023)	2023	RF, DNN, MLP, AdaBosst, KNN, LightGBM	RoEduNet- SIMARGL2021 CICIDS-2017 NSL-KDD	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Sivamohan et al. (Sivamohan et al., 2023)	2023	Bi-LSTM	CICIDS-2018 NSL-KDD	SHAP	Local, Global	Agnostic	Post-hoc
El Houda et al. (El Houda et al., 2023)	2023	Federated learning	UNSW-NB15	RuleFit, SHAP	Local, Global	Agnostic	Post-hoc
Lopes et al. (Lopes et al., 2023)	2023	ANN	LYCOS-IDS2017	SHAP	Local	Agnostic	Post-hoc
Islam et al. (Islam et al., 2023)	2023	RF, DT, ANN, CNN	UNSW-NB15	LIME, SHAP, ELI5, ProtoDash	Local, Global	Agnostic	Post-hoc
Hasan et al. (Hasan et al., 2023)	2023	Boosting algorithms	SCVIC-APT-2021	SHAP	Local	Agnostic	Post-hoc
Kostopoulos et al. (Kostopoulos et al., 2023)	2023	XGBoost, MLP	DGArchive	SHAP	Local, Global	Agnostic	Post-hoc
Thi et al. (Thi et al., 2023)	2023	GNN, CNN-GRU	NF-ToN-IoT, DARPA TCE3	SHAP	Local	Agnostic	Post-hoc
Wei et al. (Wei et al., 2023)	2023	MLP	CICDDOS2019	SHAP	Local, Global	Agnostic	Post-hoc

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Table 4 (continued)

Reference	Year	ML/DL Models	Dataset	XAI Models	Local/ Global	Model Specific/ Agnostic	Post-hoc/ Intrinsic
Irnée et al. (Irnée et al., 2023)	2023	XGBoost	CIRA-CIC-DoHBrw-2020	SHAP	Local, Global	Agnostic	Post-hoc
Rumesh et al. (Rumesh et al., 2023)	2023	DNN	UNSW-NB15 NSL-KDD	SHAP	Global	Agnostic	Post-hoc
Gebrehans et al. (Gebrehans et al., 2023)	2023	CNN	CSIRO Exfiltration Netlab DGA	SHAP	Local, Global	Agnostic	Post-hoc
Alani and Damiani (Alani and Damiani, 2023)	2023	RF, LR, DT, GNB, XGBoost	IoT-ID, TON_IoT	SHAP	Local	Agnostic	Post-hoc
Bashaiwth et al. (Bashaiwth et al., 2023)	2023	LSTM	CICIDS2017 CSECICIDS2018 CICDDoS2019	LIME, SHAP, Anchor, LORE	Local	Agnostic	Post-hoc
Feng et al. (Feng et al., 2024)	2023	KNN	DARPA 2009 DDoS CAIDA 2007 DDoS FRGP NTP Flow Data DDoS Chargin 2016 FRGP Colorado Traffic	KNN	Local, Global	Specific	Intrinsic
Miao et al. (Miao et al., 2022)	2022	CNN	CIC-IDS2017	SHAP	Local	Agnostic	Post-hoc
Zhu et al. (Zhu et al., 2022)	2022	GNN	Botnet graph dataset	Self Explainable	Global	Specific	Intrinsic
Mowla et al. (Mowla et al., 2022)	2022	KNN, MLP, RF	SAD, ORNL	LIME	Local	Agnostic	Post-hoc
Alani and Miri (Alani and Miri, 2022)	2022	DT, GNB, RF, LR	TON_IoT, IoT-23, IoT-ID	SHAP	Local	Agnostic	Post-hoc
Szczepanski et al. (Szczepanski et al., 2022)	2022	ANN	IoT-23	Oracle-Explainer	Local	Agnostic	Post-hoc
El-Mihoub et al. (El-Mihoub et al., 2022)	2022	NB, LR, DT, SVM	ASNM-NPBO	EBM	Local, Global	Agnostic	Post-hoc
Huong et al. (Huong et al., 2022)	2022	Autoencoder	SCADA dataset	SHAP	Global	Agnostic	Post-hoc
AL-Essa et al. (AL-Essa et al., 2022a)	2022	NN	MalDroid20 CIC-IDS2017	DALEX	Global	Agnostic	Post-hoc
Chawla et al. (Chawla et al., 2022)	2022	Autoencoder	Network cell trace dataset	SHAP	Local	Agnostic	Post-hoc
Carrera et al. (Carrera et al., 2022)	2022	Autoencoder	KDD99	SHAP	Local	Agnostic	Post-hoc
Sauka et al. (Sauka et al., 2022)	2022	DT, RF,SVM, MLP	NSL-KDD CIC-IDS2017 NSL-KDD	SHAP	Local, Global	Agnostic	Post-hoc
Eriksson and Grov (Eriksson and Grov, 2022)	2022	DNN	UNSW-NB15	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Roy et al. (Roy et al., 2022)	2022	DNN	CIC-IDS2017	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Roshan and Zafar (Roshan and Zafar, 2022)	2022	Autoencoder	CIC-IDS2017	KernelSHAP	Local, Global	Agnostic	Post-hoc
Tabassum et al. (Tabassum et al., 2022)	2022	DT, RF, ANN AdaBoost, XGBoost, MLP	ToN_IoT	LIME, SHAP, ELI5	Local, Global	Agnostic	Post-hoc
Ha et al. (Ha et al., 2022)	2022	LSTM based autoencoder	SCADA dataset	Gradient SHAP	Local I Global	Agnostic	Post-hoc
Aslam et al. (Aslam et al., 2022)	2022	DT, NB, RF, XGB, AB, CB	DNS dataset	SHAP, LIME	Local, Global	Agnostic	Post-hoc
Zhou et al. (Zhou et al., 2022)	2022	DT	CIC-AWS-2018	Self explainable	Local, Global	Specific	Intrinsic
Ramyavarshini et al. (Ramyavarshini et al., 2022)	2022	–	NSL-KDD	LIME, SHAP	Local, Global	Agnostic	Post-hoc
Patil et al. (Patil et al., 2022)	2022	DT, RF, SVM	CIC-IDS2017	LIME	Local	Agnostic	Post-hoc
Younisse et al. (Younisse et al., 2022)	2022	CNN	KDD99, Distilled Kitsune-2018	SHAP	Local	Agnostic	Post-hoc
Kalutharage et al. (Kalutharage et al., 2022)	2022	Autoencoder	USB-IDS	SHAP	Local, Global	Agnostic	Post-hoc
Hutchison et al. (Hutchison et al., 2022)	2022	Autoencoder	NSL-KDD, CIC2017	Self explainable	Local, Global	Specific	Intrinsic
Yilmaz and Bardak (Yilmaz and Bardak, 2022)	2022	XGBoost, CatBoost, LightGBM	CIC-IDS2017, CSECIC- IDS2018, INSDN	SHAP	Global	Agnostic	Post-hoc
Barnard et al. (Barnard et al., 2022)	2022	XGBoost AutoEncoder	NSL-KDD	SHAP	Local	Agnostic	Post-hoc
Andresini et al. (Andresini et al., 2022)	2022	Neural Network and Attention	NSL-KDD UNSW-NB15	Self Explainable	Local	Specific	Intrinsic
Houda et al. (El Houda et al., 2022a)	2022	DNN	NSL-KDD, UNSW-NB15	RuleFit, LIME, SHAP	Local, Global	Agnostic	Post-hoc
Yiwen et al. (Yiwen et al., 2022)	2022	CNN + LSTM, XGBoost	KDD99	LIME, SHAP	Local	Agnostic	Post-hoc
Zebin et al. (Zebin et al., 2022)	2022	Random Forest	CIRA-CIC-DoHBrw-2020	SHAP	Local	Agnostic	Post-hoc
Hsupeng et al. (Hsupeng et al., 2022)	2022	XGBoost	CIC-IDS2017, CSTI-10	SHAP	Local	Agnostic	Post-hoc
Le et al. (Le et al., 2022)	2022	Random Forest	NF-BoT-IoT-v2, NF-ToN-IoT-v2	SHAP	Local, Global	Agnostic	Post-hoc
Dias et al. (Dias et al., 2022)	2022	Decision Tree	–	Self Explainable	Global	Specific	Intrinsic
Oseni et al. (Oseni et al., 2022)	2022	DNN	ToN_IoT	SHAP	Local, Global	Agnostic	Post-hoc
Lundberg et al. (Lundberg et al., 2022)	2022	DNN	Survival	SHAP	Local	Agnostic	Post-hoc

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Table 4 (continued)

Reference	Year	ML/DL Models	Dataset	XAI Models	Local/ Global	Model Specific/ Agnostic	Post-hoc/ Intrinsic
El Houda et al. (El Houda et al., 2022b)	2022	DNN	UNSW-NB15	RuleFit, SHAP	Local, Global	Agnostic	Post-hoc
Ho et al. (Ho et al., 2022)	2022	DT, CNN	KDD99, CICIDS2017	SHAP	Local, Global	Agnostic	Post-hoc
Khan et al. (Khan et al., 2022)	2022	CNN, LSTM	GSP dataset	LIME	Global	Agnostic	Post-hoc
Ables et al. (Ables et al., 2022)	2022	SOM	NSL-KDD CICIDS2017	Self Explainable	Local, Global	Specific	Post-hoc
Batchu and Seetha (Batchu and Seetha, 2022)	2022	DES techniques	CICDDoS2019	SHAP, LIME	Local, Global	Agnostic	Post-hoc
Al-Essa et al. (Al-Essa et al., 2022b)	2022	DNN	MalDroid20 CICIDS2017	SHAP	Local, Global	Agnostic	Post-hoc
Liu et al. (Liu et al., 2021)	2021	DNN	UGR'16	CEM, BRCG, EDA	Local, Global	Agnostic	Post-hoc, Intrinsic
Mane and Rao (Mane and Rao, 2021)	2021	DNN	NSL-KDD	SHAP, LIME, BRCG, CEM, ProtoDash	Local, Global	Agnostic	Post-hoc, Intrinsic
Mahbooba et al. (Mahbooba et al., 2021)	2021	Decision Tree	NSL-KDD	Self Explainable	Local	Specific	Intrinsic
Wali and Khan (Wali and Khan, 2021)	2021	Random Forest	CSE-CIC IDS 2018	SHAP	Local	Agnostic	Post-hoc
Prasad et al. (Prasad et al., 2021)	2021	ML	KDD99	TCAV	Global	Agnostic	Post-hoc
Sarhan et al. (Sarhan et al., 2021a)	2021	DFF/RF	CSE-CIC-IDS2018, BoT-IoT, ve ToN-IoT	SHAP	Local	Agnostic	Post-hoc
Tcydenova et al. (Tcydenova et al., 2021)	2021	SVM	NSL-KDD	LIME	Local	Agnostic	Post-hoc
Wang et al. (Wang et al., 2021)	2021	Multiclass/Binary Classifiers	CICIDS2017	SHAP	Local, Global	Agnostic	Post-hoc
Kim et al. (Kim et al., 2021)	2021	XGBoost	NSL-KDD	SHAP	Global	Agnostic	Post-hoc
Gu et al. (Gu et al., 2021)	2021	DNN	ADFA-LD, UNM	SHAP	Local, Global	Agnostic	Post-hoc
Islam and Eberle (Islam and Eberle, 2021)	2021	RF, ET, SVM, GB, ANN, NB	CICIDS2017	Domain Knowledge Infusion	Local	Specific	Intrinsic
Andresini et al. (Andresini et al., 2021)	2021	DNN	CICIDS2017	DALEX	Local, Global	Specific	Post-hoc
Zolanvari et al. (Zolanvari et al., 2021)	2021	ANN	NSL-KDD, UNSW-NB15	TRUST	Local	Agnostic	Post-hoc
Roshan and Zafar (Roshan and Zafar, 2021)	2021	Autoencoder	CICIDS2017	SHAP	Local	Agnostic	Post-hoc
Rao and Mane (Rao and Mane, 2021)	2021	Binary classifier	NSL-KDD	SHAP, LIME	Local	Agnostic	Post-hoc
Das et al. (Das et al., 2021)	2021	RF, DT, LR	NSL-KDD	SHAP, LIME	Local, Global	Agnostic	Post-hoc
Wierzbicki et al. (Wierzbicki et al., 2021)	2021	GBM, GLM, RF, NN	RegSOC-KES2021	SHAP	Local	Agnostic	Post-hoc
Sarhan et al. (Sarhan et al., 2021b)	2021	DFF/RF	CSE-CIC-IDS2018, BoTIoT, and ToN-IoT	SHAP	Local	Agnostic	Post-hoc
Dang (Dang, 2021)	2021	XGBoost	–	PDP, SHAP	Local, Global	Agnostic	Post-hoc
Szczepanski et al. (Szczepanski et al., 2021)	2021	ANN	IoT-23	LIME Oracle-Explainer	Local	Agnostic	Post-hoc
Al-Imran et al. (Al-Imran et al., 2021)	2021	LSTM	Kyoto 2013	LIME	Local	Agnostic	Post-hoc
Caforio et al. (Caforio et al., 2021)	2021	CNN	KDD99, NSL-KDD, UNSW-NB15	GradCAM	Local	Specific	Post-hoc
Wickramasinghe et al. (Wickramasinghe et al., 2021)	2021	SOM	KDD99, DoHBrw-2020	Self Explainable	Local, Global	Specific	Post-hoc
Alenezi and Ludwig (Alenezi and Ludwig, 2021)	2021	XGBoost, RF	ISCX-URL2016, CICMalDroid 2020	SHAP	Local, Global	Agnostic	Post-hoc
AL-Essa (AL-Essa et al., 2021)	2021	XGBoost	NSL-KDD CICIDS2017	DALEX	Local, Global	Agnostic	Post-hoc
Botana et al. (Botana et al., 2020)	2020	DT	KDD99 ISCXIDS 2012	Self Explainable	Local	Specific	Ante-hoc
Rathod et al. (Rathod et al., 2020)	2020	Autoencoder	Gas Pipeline system dataset	LIME, LRP	Local, Global	Agnostic	Post-hoc
Sejr et al. (Sejr et al., 2020)	2020	KNN	ECML PKDD 2007 dataset	KNN	Local, Global	Specific	Intrinsic
Karn et al. (Karn et al., 2020)	2020	DT, EML, ANN, LSTM	Syscalls	SHAP, LIME	Local	Agnostic	Post-hoc
Wang et al. (Wang et al., 2020)	2020	Multiclass/Binary Classifiers	NSL-KDD	SHAP	Local, Global	Agnostic	Post-hoc
Reyes et al. (Reyes et al., 2020)	2020	RF/NB/XGB	AWID	SHAP	Local, Global	Agnostic	Post-hoc
Szczepanski et al. (Szczepanski et al., 2020)	2020	ANN/DT	CICIDS2017	Clustering	Local	Agnostic	Post-hoc
İslam et al. (İslam et al., 2020)	2020	RF, ET, SVM, GB, ANN, NB	CICIDS2017	Domain Knowledge Infusion	Local	Specific	Intrinsic
Antwarg et al. (Antwarg et al., 2020)	2020	Autoencoder	KDD99	SHAP	Local	Agnostic	Post-hoc
Ahn et al. (Ahn et al., 2020)	2020	CNN	ISCX VPN-nonVPN	Genetic algorithm	Local	Specific	Intrinsic
Wu et al. (Wu et al., 2020)	2020	CNN	NSL-KDD	LIME	Local	Agnostic	Post-hoc
Sarica and Angin (Sarica and Angin, 2020)	2020	RF	SDN dataset	Self Explainable	Local, Global	Specific	Intrinsic
Dang (Dang et al., 2020)	2020	XGBoost	CICIDS2017	PDP, SHAP	Local, Global	Agnostic	Post-hoc
Guerra-Manzanares et al. (Guerra-Manzanares et al., 2019)	2019	DT, RF, KNN	IoT Botnet dataset	LIME	Local	Agnostic	Post-hoc

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Table 4 (continued)

Reference	Year	ML/DL Models	Dataset	XAI Models	Local/ Global	Model Specific/ Agnostic	Post-hoc/ Intrinsic
Morichetta et al. (Morichetta et al., 2019)	2019	SVM	Youtube QoE	LIME	Local	Agnostic	Post-hoc
Li et al. (Li et al., 2019)	2019	RNN	NSL-KDD	Regression + Fused Lasso	Local	Specific	Intrinsic
Amarasinghe et al. (Amarasinghe et al., 2018)	2018	DNN	NSL-KDD	DNN Explanation Generator	Global	Specific	Post-hoc
Amarasinghe and Manic (Amarasinghe and Manic, 2018)	2018	MLP	NSL-KDD	Key Feature Extraction	Local	Agnostic	Post-hoc
Marino et al. (Marino et al., 2018)	2018	MLP	NSL-KDD	Adversarial	Local	Agnostic	Post-hoc

outdated and does not include many modern attack types and methods.

3.4.4. Other key findings in intrusion detection systems

When Table 4 is examined, it is evident that most of the studies in the literature have used model-agnostic approaches such as SHAP and LIME. However, in addition to intrinsically explainable models like DT and RF, there are various studies that employ specific explanation methods developed particularly for DL models such as CNN, DNN, and autoencoders.

Furthermore, when reviewing the literature, it is notable that the number of studies on the explainability of intrusion detection systems in IoT networks has increased in recent years, in parallel with the rapid development of IoT technologies and networks. Additionally, as clearly shown in Table 4, XAI-based IDS studies have garnered increasing attention, with more research being conducted by researchers each year.

4. Challenges, open issues and future directions

This section discusses the insights and lessons learned from examining XAI methodologies for IDSs. Moreover, based on detailed reviews in this article, we discussed open issues and possibly highlighted research challenges for future directions. In this study, XAI techniques are discussed in detail. Almost all explainability approaches in the literature are presented in this study to guide the readers and find the answers to the research questions.

While XAI presents many conveniences and challenges, it attracts increasing attention from AI researchers, users, professionals, and policymakers. With the help of XAI approaches, important steps have been taken towards “responsible AI” by ensuring trust and transparency of AI models, easy adoption and debugging of errors or ML/DL architectures. Moreover, XAI has brought some challenges as well as significant gains. Although XAI-based IDSs can be a powerful tool for their gains and impacts, there are certain challenges in applying XAI to IDSs. The main challenges are presented and discussed below under different subtitles.

i. Attacks against XAI-based IDSs

- IDSs are mainly focused on securing information systems by detecting security threats. However, since XAI-based IDSs are more transparent about their internal structures and decision mechanisms, attackers can easily detect the vulnerabilities of IDSs and perform advanced attacks specific to them.
- Attacks (such as poisoning) to manipulate or damage training data, attacks to evade authentication mechanisms (such as face authentication system), attacks to manipulate defense mechanisms, and counterfactual gaming attacks (Kuppa and Le-Khac, 2021) are recent threats.
- Serious social, economic and ethical problems may arise in misusing the explainability features of XAI-based IDSs.
- Appropriate rules and regulations being formulated to ensure the constructive use of XAI-enabled IDSs in cases of abuse can cause protected systems and safety of people affected by attacks (Ahmed et al., 2022).

ii. Privacy issues

- Even if protocol-driven to indicate users’ data under certain conditions (Srivastava et al., 2022) and privacy in design care should be taken into account and ensured that the model is not reversible (Arrieta et al., 2020), AI models need to consider privacy issues throughout the system lifecycle but more explainability in some cases make the privacy violated.

iii. The trade-off between performance and explainability

- Use XAI algorithms in practical security applications, the time complexity of the algorithms needs to be addressed (Srivastava et al., 2022) and computational resources, model accuracy, and computational power taking into account data diversity, data speed, and data quality (Arrieta et al., 2020) are trade-off between performance and explainability. Handling these issues, optimized XAI algorithms should be developed by considering multiple constraints.
- When XAI-based IDSs are evaluated, some difficulties in performance and gaining significant improvement might be encountered in explainability. Although some self-explanatory XAI approaches provide transparency and the applicants might easily understand the decision-making process, many studies do not always provide satisfactory results such as best-performing DL models are the least explainable and create worst explainability for high-performing models, especially in systems containing complex and large data with high traffic rates. So ensuring performance and explainability simultaneously is also one of the important challenges.

iv. Evaluation metrics

- These are another important factors and challenges in implementing XAI approaches in IDSs. As seen from Table 3, many XAI methods have different approaches, performances, accuracies, and measurements to evaluate the completeness of explanations.
- A set of standard explainability evaluation metrics for XAI approaches and applications were also another challenges. According to different criteria concerning abilities to reflect the facts learned from the underlying bunch of real data that in some cases penalizing and rewarding XAI methods are necessary for achieving certain levels of consistency (Elkhawaga et al., 2024).
- Developing AI techniques with explainabilities will require both high performance and more descriptive models in the future as a real challenge, especially reliable, trustable, fair, stable, consistent and user-friendly approaches or models.
- Legal infrastructures required to ensure XAI model performances with trust are also challenges not only for IDSs but also all critical fields in more concrete and universal concepts and to handle security and privacy issues with more concern.

5. Results and discussions

As summarized in this article, XAI models offers solutions to IDS problems given above by creating more interpretable and understandable AI models. XAI employs various methods to help comprehend AI

Table 5
Evaluation of XAI papers in IDS.

Perspective	Types	Papers	Count	Summary
XAI Models	LIME	(Gaspar et al., 2024), (Ahamed Maricar et al., 2024), (Sivamohan and Sridhar, 2023), (Almuqren et al., 2023), (Sharma et al., 2023), (Rehman et al., 2023), (Sharma et al., 2024) (Sevri and Karacan, 2023), (Gurbuz et al., 2023), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Muna et al., 2023), (Hassan et al., 2023), (Pawlicki et al., 2023), (Arreche et al., 2023), (Islam et al., 2023), (Bashaiwth et al., 2023), (Mowla et al., 2022), (Eriksson and Grov, 2022), (Roy et al., 2022), (Tabassum et al., 2022), (Aslam et al., 2022) (Ramyavarshini et al., 2022), (Patil et al., 2022), (El Houda et al., 2022a), (Yiwen et al., 2022), (Khan et al., 2022), (Batchu and Seetha, 2022), (Mane and Rao, 2021) (Tcydenova et al., 2021), (Rao and Mane, 2021), (Das et al., 2021), (Szczepanski et al., 2021), (Al-Imran et al., 2021), (Rathod et al., 2020), (Karn et al., 2020) (Wu et al., 2020), (Guerra-Manzanares et al., 2019), (Morichetta et al., 2019)	38	SHAP is a widely adopted XAI method in IDS studies due to its ability to provide both global and local explanations for model decisions by assigning importance scores to individual features. LIME's ability to explain individual predictions is particularly useful for security experts who need to interpret why certain examples are classified as attacks. This advanced capability has made it the most preferred method in the literature after SHAP. Decision trees do not require post-hoc explainability methods such as SHAP or LIME, as they are inherently interpretable.
	SHAP	(Gaspar et al., 2024), (Hooshmand et al., 2024), (Mallampati and Hari, 2024), (Sivamohan and Sridhar, 2023), (Marwa et al., 2023), (Lanfer et al., 2023), (Siganos et al., 2023) (Manh-Dung et al., 2023), (Sherif and Mohamed, 2023), (Youcef et al., 2023), (Pande and Khamparia, 2023), (Sharma et al., 2024), (Sevri and Karacan, 2023), (Larriva-Novo et al., 2023), (Gurbuz et al., 2023) , (Lavielle and Atlas, 2023), (Chinaechetam et al., 2023), (Recio-Garcia et al., 2023), (Kalutharage et al., 2023), (Javeed et al., 2024), (Bacevicius and Paulauskaite-Taraseviciene, 2023) (Patil et al., 2023), (Muna et al., 2023), (Pawlicki et al., 2023), (Arreche et al., 2023), (Sivamohan et al., 2023), (El Houda et al., 2023), (Lopes et al., 2023) (Islam et al., 2023), (Hasan et al., 2023), (Kostopoulos et al., 2023), (Wei et al., 2023), (Thi et al., 2023), (Irénee et al., 2023), (Rumesh et al., 2023) (Gebrehans et al., 2023), (Alani and Damiani, 2023), (Bashaiwth et al., 2023), (Miao et al., 2022), (Alani and Miri, 2022), (Huong et al., 2022), (Chawla et al., 2022) (Carrera et al., 2022), (Sauka et al., 2022), (Eriksson and Grov, 2022), (Roy et al., 2022), (Tabassum et al., 2022), (Aslam et al., 2022), (Ramyavarshini et al., 2022) (Younisse et al., 2022), (Kalutharage et al., 2022), (Yilmaz and Bardak, 2022), (Barnard et al., 2022), (El Houda et al., 2022a), (Yiwen et al., 2022), (Zebin et al., 2022) (Hsupeng et al., 2022), (Le et al., 2022), (Oseni et al., 2022), (Lundberg et al., 2022), (El Houda et al., 2022b), (Ho et al., 2022), (Batchu and Seetha, 2022) (AL-Essa et al., 2022b), (Mane and Rao, 2021), (Wali and Khan, 2021), (Sarhan et al., 2021a), (Wang et al., 2021), (Kim et al., 2021), (Gu et al., 2021) (Roshan and Zafar, 2021), (Rao and Mane, 2021), (Das et al., 2021), (Wierzbicki et al., 2021), (Sarhan et al., 2021b), (Dang, 2021), (Alenezi and Ludwig, 2021) (Karn et al., 2020), (Wang et al., 2020), (Reyes et al., 2020), (Antwarg et al., 2020), (Dang et al., 2020)	82	
	ICE	Marwa et al. (2023)	1	
	PDP	(Marwa et al., 2023), (Dang, 2021), (Dang et al., 2020)	3	
	PFI	Marwa et al. (2023)	1	
	IG	(Gurbuz et al., 2023)	1	
	FI, PI	(Recio-Garcia et al., 2023)	1	
	RuleFit	(El Houda et al., 2023), (El Houda et al., 2022a), (El Houda et al., 2022b)	3	
	Anchor, LORE	Bashaiwth et al. (2023)	1	
	LRP	Rathod et al. (2020)	1	
	KNN	(Sejr et al., 2020), (Feng et al., 2024)	2	
	Oracle-Explainer	(Szczepanski et al., 2022), (Szczepanski et al., 2021)	2	
	KernelSHAP	Roshan and Zafar (2022)	1	
	Gradient SHAP	Ha et al. (2022)	1	
	CEM, BRGC	(Liu et al., 2021), (Mane and Rao, 2021)	2	
	TCAV	Prasad et al. (2021)	1	
	TRUST	Zolanvari et al. (2021)	1	
	GradCAM	Caforio et al. (2021)	1	
	Domain Knowledge	(Islam et al., 2020), (Islam and Eberle, 2021)	2	
	Infusion			
	Clustering	Szczepanski et al. (2020)	1	
	Genetic algorithm	Ahn et al. (2020)	1	

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Table 5 (continued)

Perspective	Types	Papers	Count	Summary		
Model Specific	Regression + Fused Lasso	Li et al. (2019)	1	Most of the studies in the literature have used model-agnostic approaches such as SHAP and LIME. However, in addition to intrinsically explainable models like DT and RF, there are several studies that use specific explanation methods developed specifically for DL models such as CNN, DNN, and autoencoders.		
	DNN Explanation Generator	Amarasinghe et al. (2018)	1			
	Key Feature Extraction	Amarasinghe and Manic (2018)	1			
	Adversarial	Marino et al. (2018)	1			
	Self Explainable	(Lanvin et al., 2023), (Zhou et al., 2023), (Zhu et al., 2022), (Zhou et al., 2022), (Hutchison et al., 2022), (Andresini et al., 2022) (Dias et al., 2022), (Ables et al., 2022), (Mahbooba et al., 2021), (Wickramasinghe et al., 2021), (Botana et al., 2020), (Sarica and Angin, 2020)	12			
		(Lanvin et al., 2023), (Feng et al., 2024), (Zhu et al., 2022), (Zhou et al., 2022), (Hutchison et al., 2022), (Andresini et al., 2022) (Dias et al., 2022), (Ables et al., 2022), (Mahbooba et al., 2021), (Islam and Eberle, 2021), (Andresini et al., 2021), (Caforio et al., 2021) (Wickramasinghe et al., 2021), (Botana et al., 2020), (Sejr et al., 2020), (Islam et al., 2020), (Ahn et al., 2020), (Sarica and Angin, 2020)	20			
	Intrinsic Explanation		(Li et al., 2019), (Amarasinghe et al., 2018)		16	Most of the studies in the literature have used previously developed XAI methods to make the results explainable, and as a result, the explanations in most of the studies are post-hoc.
			(Lanvin et al., 2023), (Feng et al., 2024), (Zhu et al., 2022), (Rehman et al., 2023), (Zhou et al., 2022), (Andresini et al., 2022) (Dias et al., 2022), (Liu et al., 2021), (Mane and Rao, 2021), (Mahbooba et al., 2021), (Islam and Eberle, 2021), (Sejr et al., 2020) (Islam et al., 2020), (Ahn et al., 2020), (Sarica and Angin, 2020), (Li et al., 2019)			
	Dataset	NSL-KDD	(Hooshmand et al., 2024), (Sivamohan and Sridhar, 2023), (Marwa et al., 2023), (Lanfer et al., 2023), (Almuqren et al., 2023), (Sharma et al., 2023) (Rehman et al., 2023), (Yousef et al., 2023), (Pande and Khamparia, 2023), (Sharma et al., 2024), (Arreche et al., 2023), (Sivamohan et al., 2023) (Rumesh et al., 2023), (Carrera et al., 2022), (Sauka et al., 2022), (Ramyavarshini et al., 2022), (Hutchison et al., 2022), (Barnard et al., 2022) (Andresini et al., 2022), (El Houda et al., 2022a), (Ables et al., 2022), (Mane and Rao, 2021), (Mahbooba et al., 2021), (Tcydenova et al., 2021) (Kim et al., 2021), (Zolanvari et al., 2021), (Rao and Mane, 2021), (Das et al., 2021), (Caforio et al., 2021), (AL-Essa et al., 2021) (Wang et al., 2020), (Wu et al., 2020), (Li et al., 2019), (Amarasinghe et al., 2018), (Amarasinghe and Manic, 2018), (Marino et al., 2018)		36	The NSL-KDD dataset has been preferred more often because it is an older dataset, and its structure is suitable for many AI models. The NSL-KDD dataset is followed by the UNSW-NB15 and CIC-IDS2017 datasets, which have become increasingly popular among researchers in recent years. It is notable that the KDD99 dataset is no longer preferred in many studies, despite being the oldest dataset. The reason for this is that it has become obsolete and does not include many modern attack types and methods.
		UNSW-NB15	(Hooshmand et al., 2024), (Mallampati and Hari, 2024), (Marwa et al., 2023), (Pande and Khamparia, 2023), (Sharma et al., 2024), (Larriva-Novo et al., 2023), (El Houda et al., 2023), (Islam et al., 2023), (Rumesh et al., 2023), (Eriksson and Grov, 2022), (Andresini et al., 2022), (El Houda et al., 2022a), (El Houda et al., 2022b), (Zolanvari et al., 2021), (Caforio et al., 2021)		15	
CIC-IDS2017		(Ahamed Maricar et al., 2024), (Mallampati and Hari, 2024), (Almuqren et al., 2023), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Lanvin et al., 2023), (Miao et al., 2022), (AL-Essa et al., 2022a), (Carrera et al., 2022), (Roy et al., 2022), (Roshan and Zafar, 2022), (Patil et al., 2022), (Yilmaz and Bardak, 2022), (Hsupeng et al., 2022)	13			
ADFA-LD		(Gu et al., 2021), (Gaspar et al., 2024)	2			
Honeypot		Sivamohan and Sridhar (2023)	1			
TON_IoT		(Marwa et al., 2023), (Alani and Damiani, 2023), (Alani and Miri, 2022), (Tabassum et al., 2022), (Oseni et al., 2022)	5			
CIC-IoT-Dataset-2022, IEC 69870-5-104		Siganos et al. (2023)	1			
CSE-CIC-IDS2018		(Manh-Dung et al., 2023), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Bashaiwth et al., 2023), (Yilmaz and Bardak, 2022), (Sarhan et al., 2021a), (Sarhan et al., 2021b)	6			
CIC-AWS-2018		(Zhou et al., 2023), (Zhou et al., 2022)	2			
CPS Dataset		(Sherif and Mohamed, 2023), (Patil et al., 2023)	2			
Intensive Care Unit (ICU)		(Gurbuz et al., 2023)	1			
Microsoft Defender's logs		Lavieille and Atlas, 2023	1			
Edge-IIoTset		Chinaechetam et al. (2023)	1			
SQL injection dataset		(Recio-Garcia et al., 2023)	1			

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Table 5 (continued)

Perspective	Types	Papers	Count	Summary
AI Models	USB-IDS	(Kalutharage et al., 2023), (Kalutharage et al., 2022)	2	Decision tree and random forest algorithms have been widely preferred in XAI-supported IDS studies. Decision trees and random forests are among the most interpretable AI models available in machine learning. For this reason, these methods have been frequently used in XAI studies. In addition, these algorithms are commonly used to evaluate prediction performance in IDS studies alongside traditional ML algorithms such as SVM, MLP, NB, KNN, and LR. Following traditional ML algorithms, deep learning-based techniques such as ANN, DNN, CNN, LSTM, GBM, and Autoencoder have been increasingly preferred in XAI-supported IDS systems, especially in recent years. These techniques are favored in many IDS systems because they offer significantly better predictive performance compared to traditional ML techniques.
	CICDoS2019	(Batchu and Seetha, 2022), (Thi et al., 2023), (Bashaiwth et al., 2023)	3	
	IoTID20	Muna et al. (2023)	1	
	DoS dataset	Hassan et al. (2023)	1	
	DDoS Backscatter	Pawlicki et al. (2023)	1	
	RoEduNet-	Arreche et al. (2023)	1	
	SIMARGL2021			
	LYCOS-IDS2017	Lopes et al. (2023)	1	
	SCVIC-APT-2021	Hasan et al. (2023)	1	
	DGArchive	Kostopoulos et al. (2023)	1	
	NF-ToN-IoT, DARPA	Thi et al. (2023)	1	
	TCE3			
	CIRA-CIC-DoHBrw-2020	(Zebin et al., 2022), (Irénee et al., 2023)	2	
	CSIRO Exfiltration	Gebrehans et al. (2023)	1	
	Netlab DGA			
	IoT-ID	(Alani and Damiani, 2023), (Alani and Miri, 2022)	2	
	IoT-23	(Szczepanski et al., 2021), (Szczepanski et al., 2022), (Alani and Miri, 2022)	3	
	DARPA 2009 DDoS	Feng et al. (2024)	1	
	CAIDA 2007 DDoS			
	FRGP NTP Flow Data			
	FRGP Colorado			
	Traffic			
	DDoS Chargin 2016			
	Botnet graph dataset	Zhu et al. (2022)	1	
	SAD, ORNL	Mowla et al. (2022)	1	
	ASNM-NPBO	El-Mihoub et al. (2022)	1	
	SCADA dataset	Ha et al. (2022)	1	
	MalDroid20	(AL-Essa et al., 2022b), (AL-Essa et al., 2022a)	2	
	Network cell trace dataset	Chawla et al. (2022)	1	
	DNS dataset	Aslam et al. (2022)	1	
	KDD99	(Yiwen et al., 2022), (Younisse et al., 2022), (Ho et al., 2022), (Prasad et al., 2021), (Caforio et al., 2021), (Wickramasinghe et al., 2021)	9	
		(Botana et al., 2020), (Antwarg et al., 2020), (Carrera et al., 2022)		
	Distilled Kitsune-2018	Younisse et al. (2022)	1	
	CSTI-10	Hsupeng et al. (2022)	1	
	NF-BoT-IoT-v2,	Le et al. (2022)	1	
	NF-ToN-IoT-v2			
	Survival	Lundberg et al. (2022)	1	
	GSP dataset	Khan et al. (2022)	1	
	UGR'16	Liu et al. (2021)	1	
	RegSOC-KES2021	Wierzbicki et al. (2021)	1	
	Kyoto 2013	Al-Imran et al. (2021)	1	
	ISCX-URL2016	Alenezi and Ludwig (2021)	1	
	ISCXIDS 2012	Botana et al. (2020)	1	
	ISCX VPN-nonVPN	Ahn et al. (2020)	1	
	Gas Pipeline system dataset	Rathod et al. (2020)	1	
	ECML PKDD 2007 dataset	Sejr et al. (2020)	1	
	AWID	Reyes et al. (2020)	1	
	SDN dataset	Sarica and Angin (2020)	1	
	IoT Botnet dataset	Guerra-Manzanares et al. (2019)	1	
	Youtube QoE	Morichetta et al. (2019)	1	
	CNN	(Manh-Dung et al., 2023), (Sherif and Mohamed, 2023), (Rehman et al., 2023), (Sharma et al., 2024), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Islam et al., 2023), (Gebrehans et al., 2023), (Miao et al., 2022), (Younisse et al., 2022), (Ho et al., 2022), (Khan et al., 2022), (Caforio et al., 2021), (Ahn et al., 2020), (Wu et al., 2020)	14	
	LSTM	(Marwa et al., 2023), (Youcef et al., 2023), (Bashaiwth et al., 2023), (Khan et al., 2022), (Al-Imran et al., 2021), (Karn et al., 2020)	6	
	DNN	(Sharma et al., 2023), (Pande and Khamparia, 2023), (Sharma et al., 2024), (Sevri and Karacan, 2023), (Pawlicki et al., 2023), (Arreche et al., 2023), (Rumesh et al., 2023), (Eriksson and Grov, 2022), (Roy et al., 2022), (El Houda et al., 2022a), (Oseni et al., 2022), (Lundberg et al., 2022), (El Houda et al., 2022b), (AL-Essa et al., 2022b), (Liu et al., 2021), (Mane and Rao, 2021), (Gu et al., 2021), (Andresini et al., 2021), (Amarasinghe et al., 2018)	19	

(continued on next page)

Table 5 (continued)

Perspective	Types	Papers	Count	Summary
	XGBoost	(Kostaspanis et al., 2023), (Tabassum et al., 2022), (2024), (and Eberle, 2023), (Zahid, 2023), (Murat, 2021), (2023), (Kostopoulou, 2021), (2023), (Ali, 2020), (2023), (Sherif and Mohamed, 2023), (Tad, 2023), (Gurbuz, 2022), (2023), (Bardak, 2022), (Bakir, 2023), (2023), (Lopez, 2022), (2023), (Islam et al., 2020), (2023), (Patil et al., 2021), (2022a), (Wierzbicki et al., 2021), (Alenezi and Ludwig, 2021), (AL-Essa et al., 2021), (Dang et al., 2020)	13	
	RF	(Lanfer et al., 2023), (Sherif and Mohamed, 2023), (Larriva-Novo et al., 2023), (Gurbuz et al., 2023), (Recio-Garcia et al., 2023), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Hassan et al., 2023), (Arreche et al., 2023), (Islam et al., 2023), (Alani and Damiani, 2023), (Alani and Miri, 2022), (Mowla et al., 2022), (Sauka et al., 2022), (Tabassum et al., 2022), (Aslam et al., 2022), (Patil et al., 2022), (Sarhan et al., 2021a), (Islam and Eberle, 2021), (Das et al., 2021), (Wierzbicki et al., 2021), (Sarhan et al., 2021b), (Alenezi and Ludwig, 2021), (Reyes et al., 2020), (Islam et al., 2020), (Sarica and Angin, 2020), (Guerra-Manzanares et al., 2019), (Zebin et al., 2022), (Wali and Khan, 2021), (Le et al., 2022)	29	
	DT	(Larriva-Novo et al., 2023), (Gurbuz et al., 2023), (Recio-Garcia et al., 2023), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Patil et al., 2023), (Zhou et al., 2023), (Islam et al., 2023), (Alani and Damiani, 2023), (El-Mihoub et al., 2022), (Alani and Miri, 2022), (Sauka et al., 2022), (Tabassum et al., 2022), (Aslam et al., 2022), (Zhou et al., 2022), (Patil et al., 2022), (Ho et al., 2022), (Das et al., 2021), (Botana et al., 2020), (Karn et al., 2020), (Szczepanski et al., 2020), (Guerra-Manzanares et al., 2019), (Dias et al., 2022), (Mahbooba et al., 2021)	23	
	MLP	(Gaspar et al., 2024), (Sherif and Mohamed, 2023), (Larriva-Novo et al., 2023), (Gurbuz et al., 2023), (Recio-Garcia et al., 2023), (Arreche et al., 2023), (Kostopoulos et al., 2023), (Thi et al., 2023), (Mowla et al., 2022), (Sauka et al., 2022), (Tabassum et al., 2022), (Amarasinghe and Manic, 2018), (Marino et al., 2018)	13	
	KNN	(Larriva-Novo et al., 2023), (Gurbuz et al., 2023), (Patil et al., 2023), (Arreche et al., 2023), (Feng et al., 2024), (Mowla et al., 2022), (Sejr et al., 2020), (Guerra-Manzanares et al., 2019)	8	
	SVM	(Sherif and Mohamed, 2023), (Gurbuz et al., 2023), (Lavieille and Atlas, 2023), (Patil et al., 2023), (El-Mihoub et al., 2022), (Sauka et al., 2022), (Patil et al., 2022), (Tcydenova et al., 2021), (Islam and Eberle, 2021), (Islam et al., 2020), (Morichetta et al., 2019)	11	
	LR	(Larriva-Novo et al., 2023), (Recio-Garcia et al., 2023), (Bacevicius and Paulauskaite-Taraseviciene, 2023), (Alani and Damiani, 2023), (El-Mihoub et al., 2022), (Alani and Miri, 2022), (Das et al., 2021)	7	
	BiLSTM	(Sivamohan et al., 2023), (Javeed et al., 2024), (Sivamohan and Sridhar, 2023)	3	
	Bi-GRU	Javeed et al. (2024)	1	
	NB	(Sherif and Mohamed, 2023), (Gurbuz et al., 2023), (Recio-Garcia et al., 2023), (El-Mihoub et al., 2022), (Aslam et al., 2022), (Islam and Eberle, 2021), (Reyes et al., 2020), (Islam et al., 2020)	8	
	SAE	Manh-Dung et al. (2023)	1	
	Autoencoder	(Kalutharage et al., 2023), (Lanvin et al., 2023), (Carrera et al., 2022), (Chawla et al., 2022), (Huong et al., 2022), (Roshan and Zafar, 2022), (Kalutharage et al., 2022), (Hutchison et al., 2022), (Barnard et al., 2022), (Roshan and Zafar, 2021), (Rathod et al., 2020), (Antwarg et al., 2020), (Lavieille and Atlas, 2023)	13	
	LGBost	Chinaechetam et al. (2023)	1	
	AdaBoost	(Tabassum et al., 2022), (Recio-Garcia et al., 2023)	2	
	CatBoost	Yilmaz and Bardak (2022)	1	
	LightGBM	(Yilmaz and Bardak, 2022), (Arreche et al., 2023), (Mallampati and Hari, 2024)	3	
	IENN-based classification	Almugren et al. (2023)	1	
	Isolation Forest	Lavieille and Atlas (2023)	1	
	Federated learning	El Houda et al. (2023)	1	
	GNN	(Thi et al., 2023), (Zhu et al., 2022)	2	
	LSTM based autoencoder	Ha et al. (2022)	1	
	Neural Network and Attention	Andresini et al. (2022)	1	
	CNN + LSTM	Yiwen et al. (2022)	1	
	CNN + GRU	Thi et al. (2023)	1	
	DES techniques	Batchu and Seetha (2022)	1	
	RNN	Li et al. (2019)	1	
	SOM	(Ables et al., 2022), (Wickramasinghe et al., 2021)	2	

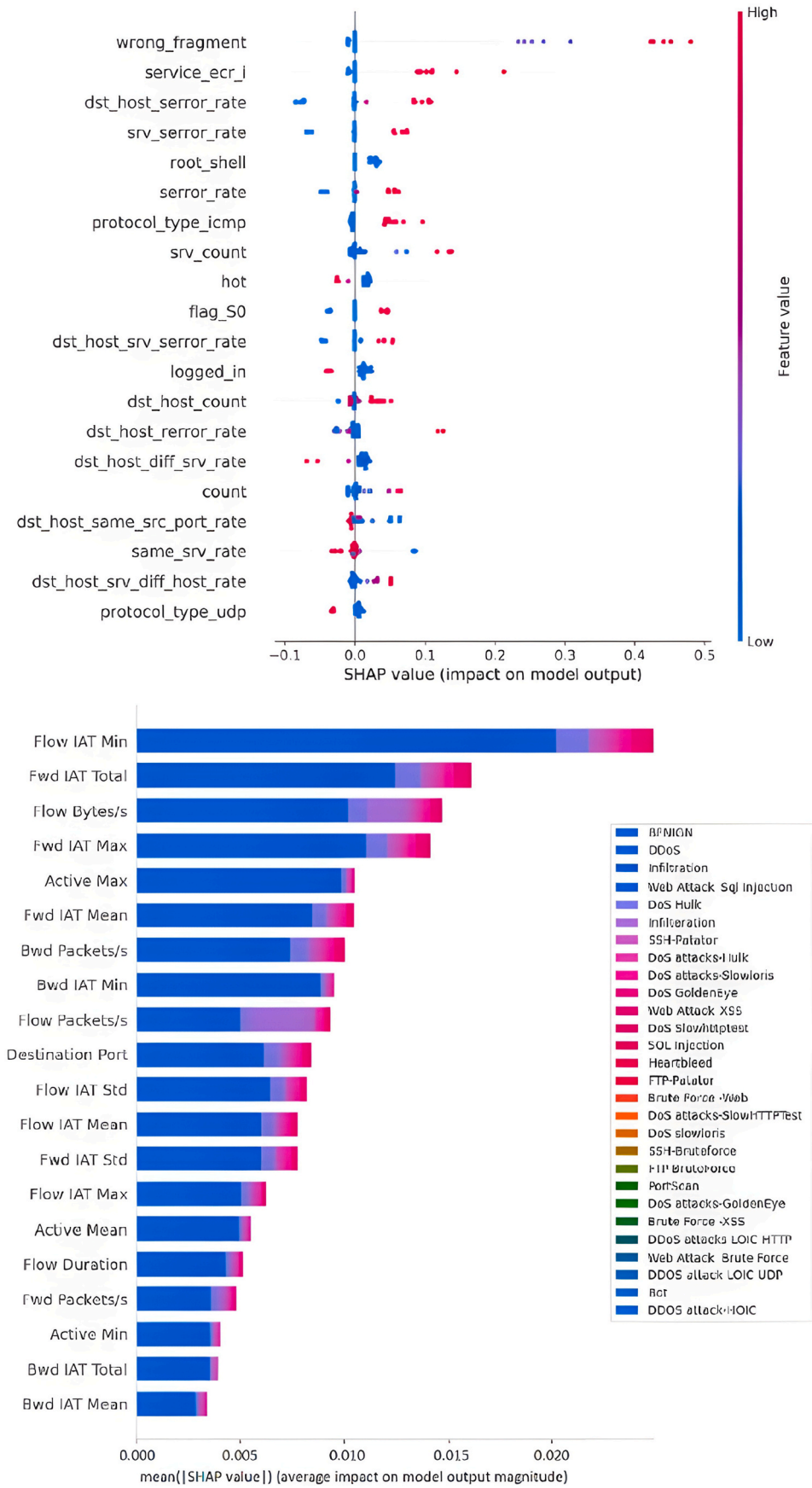


Fig. 11. SHAP explainability examples (Wang et al., 2020; Bacevicius and Paulauskaite-Taraseviciene, 2023).

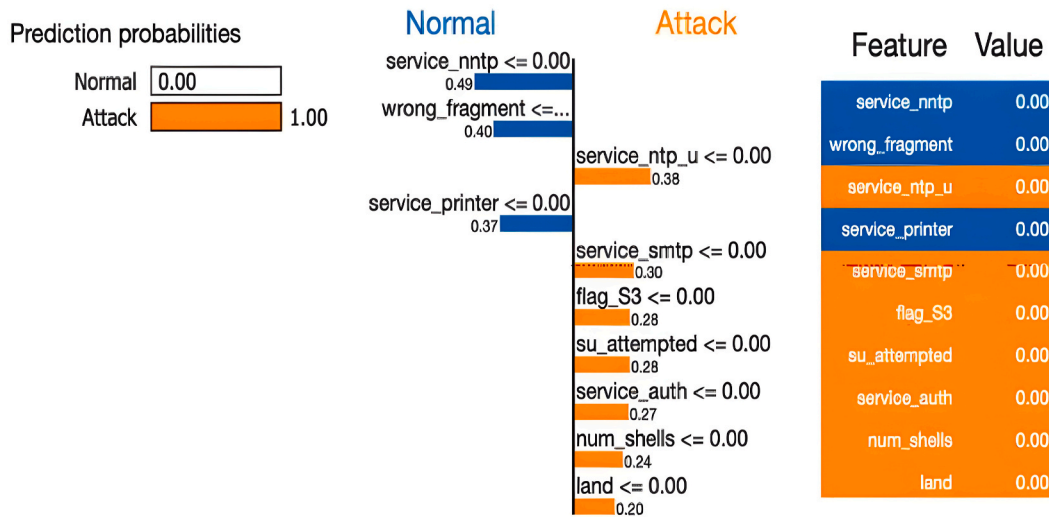


Fig. 12. LIME explainability example (Rao and Mane, 2021).

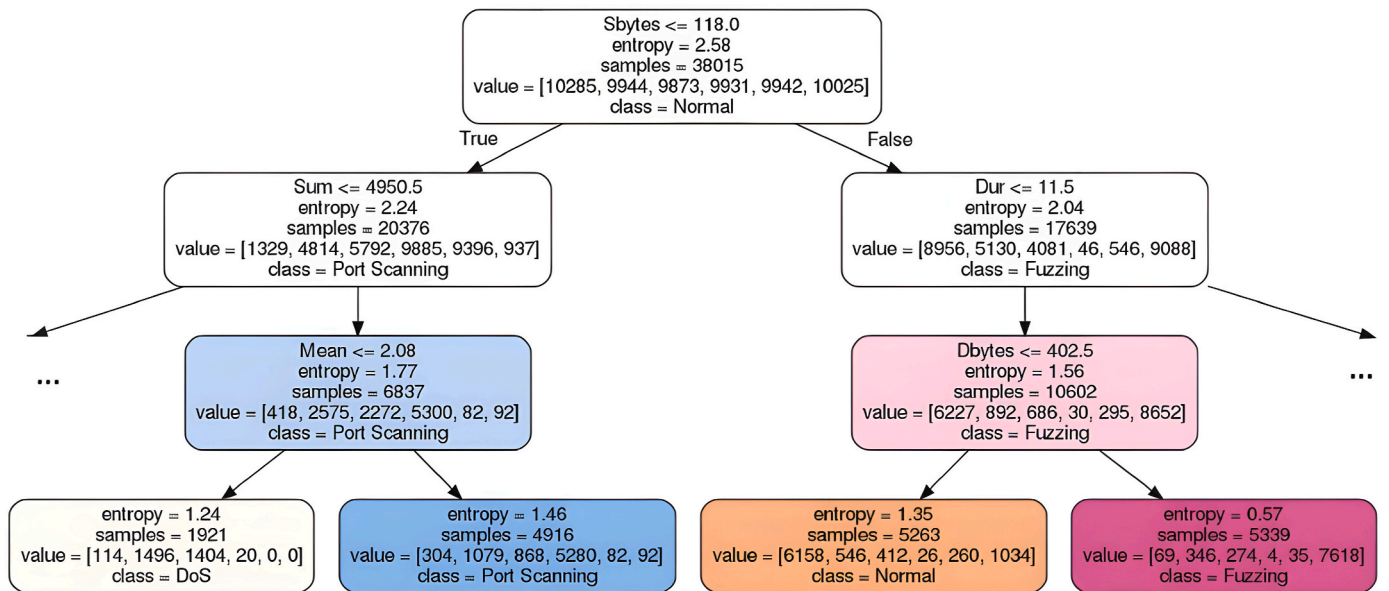


Fig. 13. An explainability example for Decision Tree (Sarica and Angin, 2020).

decisions, showing which data or data regions influenced these decisions. Despite the attention and application of XAI in many fields, its use in IDSs remains limited. IDSs, crucial for detecting data breaches and attacks, need high performance and the ability to justify their decisions. Applying XAI methods in IDSs would involve explaining decisions, analyzing results, and converting them into understandable formats. Almost all of the techniques in the literature for XAI-enabled IDSs are post-hoc and model-agnostic. This is due to the addition of existing XAI techniques as an additional module to IDSs. While developing explainability approaches specific to IDSs will significantly contribute to the cybersecurity ecosystem more than ever, these approaches might bring new vulnerabilities.

As a result of this article, XAI models have several important issues to be considered and discussed as.

- i. Explainability may be considered optional today, but with the growth of AI models and decision-making systems, it will undoubtedly be required in the future to address a number of

procedures for explaining, designing, securing, anonymizing, implementing, validating, testing with certain criteria.

- ii. Building trust in AI systems through explainability is critical but it is not always possible, especially, applicants need to understand and accept the explanations provided by XAI approaches.
- iii. Making black-box models transparent like DL or ML models is particularly challenging due to their complex and non-linear nature. Approaches like SHAP, LIME, ELI5, etc. help but they are not always sufficient to fully supportive for the decision-making process of these models.
- iv. Integrating XAI with real-time applications especially IDSs is one of the recent challenges that allows for transparent and immediate decision-making for attacks. Various XAI techniques are available in the literature (as can be seen in Table 4). This shows that LIME and SHAP are the main techniques to be used in IDSs recently.
- v. Addressing XAI approaches and related issues recently requires a multidisciplinary approach involving advancements in AI, technology, data visualization tools, data exploration platforms, user

graphical interfaces, other applications, and ethical considerations. So more attention should be paid for this approach for developing new models.

- vi. Designing explainable models for each field of the applications is crucial. Developing proper applications requires specific domain knowledge, so common XAI platforms must be developed and made public for researchers.
- vii. Even if there have been several studies available in the literature, normalized or standardized metrics and benchmarks are missing and required to evaluate the effectiveness of XAI and develop better methods. Ensuring that AI models comply with explainability adds another model that extends complexity and cost. Incorporating XAI into existing AI systems might be challenging as complex models like DL or ML tend to be more accurate but less interpretable, while simpler models are easier to explain but may not perform very well.
- viii. In general, explanations in IDSs might help users to understand patterns, detect anomalies, classify attacks, design better and intelligent systems especially XAI-based IDSs. When IDS studies in the literature are examined, using both local and global explanation techniques together provides more detailed explanations to users and helps to develop better and more reliable IDSs.
- ix. Many IDS models have not been used in practical applications because their results do not provide explanations enabling cybersecurity experts to revise their decisions. XAI might provide security professionals with basic reasoning and threat detection capabilities to reduce the risk of violations and improve overall security starting from IDSs.
- x. Even if LLMs provide many facilities to explain many decisions such as explaining IDS alerts to non-experts, they have some difficulties in terms of explainability due to their extraordinary abilities and complex structure. Ensuring transparency in the decision-making processes of LLMs and providing clear explanations for the outputs produced is important in establishing trust. In parallel with the development of LLMs, existing XAI methods cannot be directly applied due to the complex structures of LLMs, making it necessary to develop new and domain-specific explainability approaches.

6. Conclusion and future directions

This study reviews XAI models and techniques from different perspectives, examining their applications and performances in IDSs. It provides detailed definitions and terminology, explores opportunities and challenges, and summarizes approaches, tools, datasets, and technologies for developing XAI models and applications, and also highlights the necessary steps for implementing XAI in AI-based IDSs and addresses the associated risks.

The results achieved in this study given in Tables 1–5 show that the studies collectively demonstrate significant advancements in the field of IDS and cybersecurity utilizing various ML and DL models enhanced by many XAI methods. Researchers mainly focused on developing XAI techniques that are specifically designed for IDS applications; paying attention of using XAI to improve the robustness of IDS against adversarial attacks; trying to implement XAI approaches with the help of available datasets for the first time; testing available DL/ML and XAI models; suggesting new metrics for successful evaluations, etc. Combining XAI with feature selection techniques can improve model performance and interpretability (Sherif and Mohamed, 2023; Dang, 2021; Gurbuz et al., 2023). These studies demonstrate the application of various AI and ML techniques in cybersecurity, particularly IDS, and the use of XAI methods to enhance interpretability and transparency. Key methodologies include CNNs, LSTM, KNN, ensemble models, and various feature selection techniques, with explanation tools like LIME, SHAP, RuleFit and others to explain model decisions (Khan et al., 2022;

Ables et al., 2022; Song et al., 2011; Almuqren et al., 2023). This enhances transparency and understanding of how these models identify attacks (Ables et al., 2022; Siganos et al., 2023; Sharma et al., 2023; Hutchison et al., 2022; Pande and Khamparia, 2023) are pivotal in providing insights into model decisions. SHAP provides insights into how different features contribute to a model's decision, while LIME explains individual predictions by creating a simpler model around a specific data point. Many studies leverage XAI techniques to explain ML models' decision-making process for IDS.

The work can be concluded based on extending five contributions in Section 1.4 that.

- A variety of benchmark datasets (CICIDS2017, UNSW-NB15, NSL-KDD, etc.) were used to evaluate the performance of the proposed IDS models.
- The contributions summarized here highlight the importance of combining powerful ML/DL models with XAI techniques to enhance IDSs' security, reliability, and interpretability in modern cyber environments.
- Sophisticated models like autoencoders, CNN, RNN, and ensemble models were available to improve IDS or other detection and classification of cyber threats. Deep learning architectures (such as CNNs, LSTMs, Autoencoders) were employed for intrusion detection, demonstrating effectiveness in extracting features and classifying network traffic (Khan et al., 2022; Marwa et al., 2023; Kalutharage et al., 2022; Sharma et al., 2024). Especially, LIME and SHAP models effectively were used for explained models using Random Forest, XGBoost, LSTM, CNN, and Gradient Boosting ML/DL models.
- Researchers were exploring different types of ML models for IDS, including ML, RNNs, and ensemble methods. The choice of model depends on the specific requirements of the IDS, such as the type of attacks it needs to detect and the computational resources available. Explainable IDS can help to improve security by providing insights into how attacks work and how to better defend against them. Security analysts can use these explanations to identify network vulnerabilities and develop more effective security policies. XAI aided in detecting various attacks, including DDoS, DoS, zero-day web attacks, and malware (Zhou et al., 2022; Chinaechetam et al., 2023; Hassan et al., 2023).
- Implementing feature selection techniques to identify the most relevant features contributing to model predictions and aiding in the models' interpretability, explainability, and efficiency, are still an important problem to be considered. Aided in selecting the most effective features for IDS models are also possible (Aslam et al., 2022; Alani and Miri, 2022).
- The development of real-time, scalable, and robust IDS solutions is necessary for the future. Adding an explainability feature to IDS system is a real challenge to effectively handle large-scale network traffic and complex cyber-attacks. Moreover, the application of these advanced models and XAI methods across various domains, including IoT, IIoT, cyber intelligence, vehicular networks, etc. are also other directions for their versatility and effectiveness.
- Feature selection techniques were investigated to improve model interpretability and efficiency by identifying the most important features for attack detection (Batchu and Seetha, 2022; Dang, 2021; Sharma et al., 2024; Guerra-Manzanares et al., 2019). XAI methods provide insights into feature importance, helping analysts understand why a specific attack was detected (Kalutharage et al., 2022; Sharma et al., 2024; Ramyavarshini et al., 2022).
- Combining XAI with feature selection techniques can improve model performance and interpretability (Sherif and Mohamed, 2023; Dang, 2021; Gurbuz et al., 2023).
- DL models with XAI explanations were explored for anomaly detection in network traffic and SCADA systems (Sevri and Karacan, 2023; Gurbuz et al., 2023; Caforio et al., 2021). Explainable Boosting Machines were explored as an alternative for interpretable attack

detection (El-Mihoub et al., 2022). Also, Explainable Federated Learning with XAI was proposed for enhancing security and privacy in IDS (El Houda et al., 2023; Thi et al., 2023).

- Studies applied XAI techniques to IDS were designed for IoT networks and Cyber-Physical Systems (Marwa et al., 2023; Kalutharage et al., 2022; Patil et al., 2023) and IIoT, CPS, SDN and traditional networks (Marwa et al., 2023; Youcef et al., 2023; Karn et al., 2020; Patil et al., 2023)
- Increased trust and adoption of ML, DL or GenAI-based IDS might be designed and developed (Rehman et al., 2023; Islam et al., 2023; Thi et al., 2023).
- Reduced false positives and improved detection accuracy might be achieved (Hassan et al., 2023; Mowla et al., 2022; Lopes et al., 2023).

Finally, future challenge direction are.

- Privacy and security issues of GenAI models and datasets under adversarial attacks are new challenges.
- Building up new models, tuning, attention mechanism and data of AI models are new research areas to increase explainability. Dataset quality and size have become more important to develop better AI and XAI models. XAI techniques are being increasingly applied to IDSs to improve transparency and trust.
- Traditional ML models can be like black-boxes, making it difficult to understand why they make certain decisions. Developing better IDS with the help of XAI approach for complex DL models will be new direction (Sauka et al., 2022).
- Combining XAI with adversarial training to improve the robustness of IDS models against adversarial attacks will continue to be a promising research direction (AL-Essa et al., 2022b; AL-Essa et al., 2022a).
- Challenges are still associated with developing explainable IDS considering the trade-off between accuracy and explainability. Balancing the explainability and accuracy of IDS models remains an ongoing challenge (Szczepanski et al., 2021). Research in this area will be ongoing, and promising for the future.
- Security analysts and designers might benefit from improved results and decisions of understanding of IDS decisions.
- New and compact AI and XAI models, datasets, platforms, applications are always required for developing advanced detection techniques.
- The number of XAI methods is drastically increasing day by day. Especially recent developments in GenAI might bring new perspectives in developing new XAI techniques. AI, ML, DL, XAI, and especially GenAI will bring new perspectives not only to IDSs but also the other cyber security issues as well.

CRedit authorship contribution statement

Samed AL: Visualization, Methodology, Investigation, Conceptualization. **Seref Sagioglu:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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